

FACULTÉ DES SCIENCES ET TECHNOLOGIES (FST)

TD No 3-Administration Système

Devoir rédigé par : Ruth-Jose JEROME

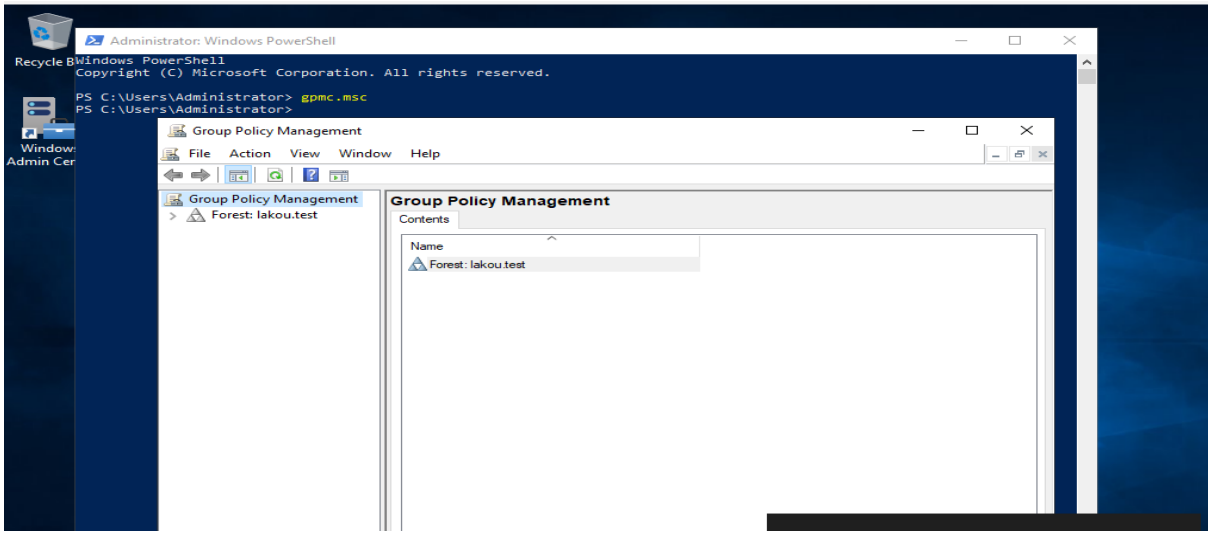
Étudiant en L4

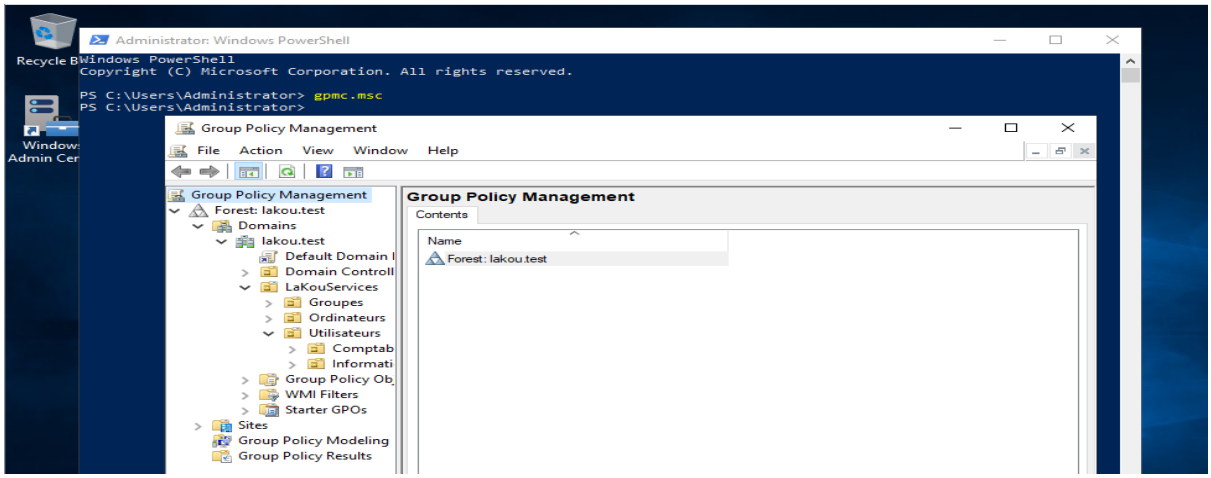
Soumis au professeur : Ismael SAINT AMOUR

DATE: 06 octobre 2026

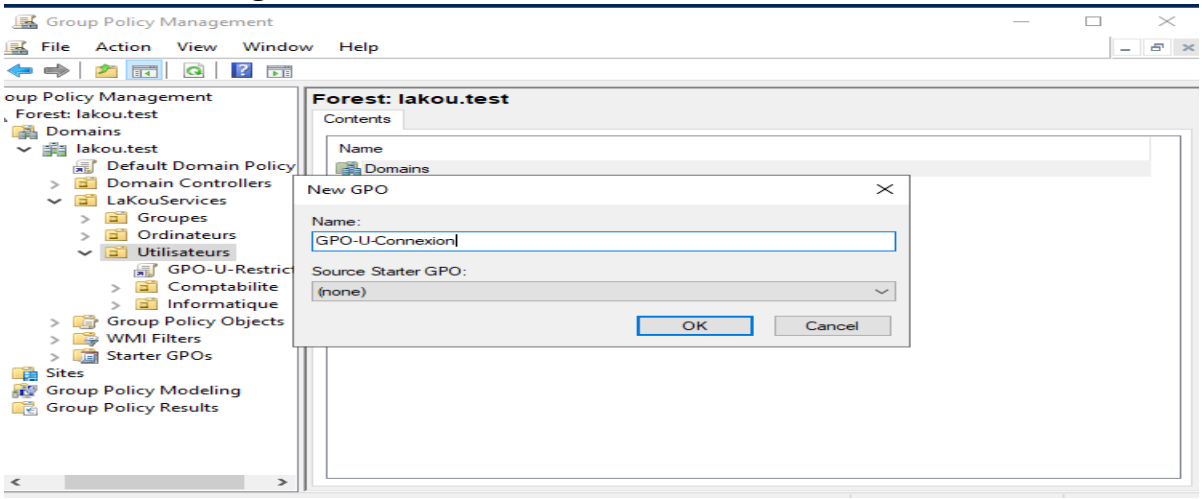
Activité 1: Créer et lier les GPO

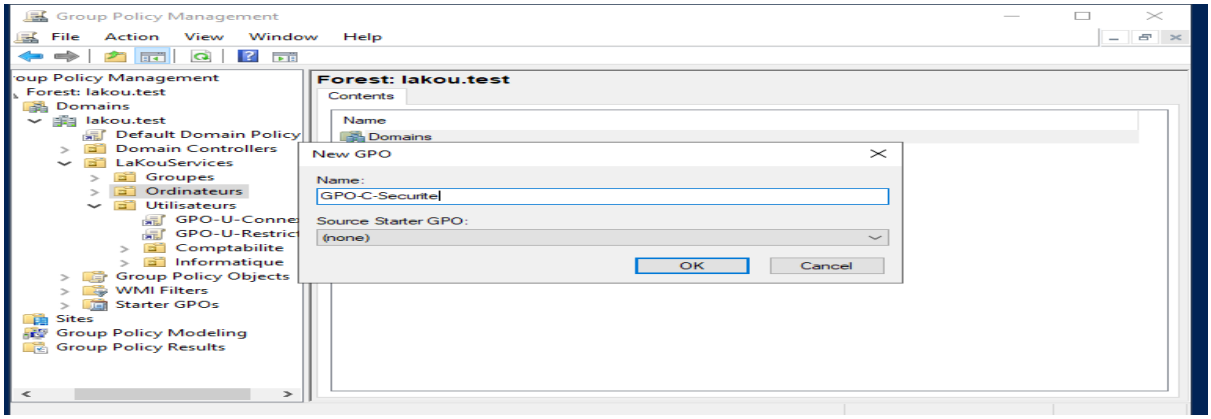
A. Ouvrir la console

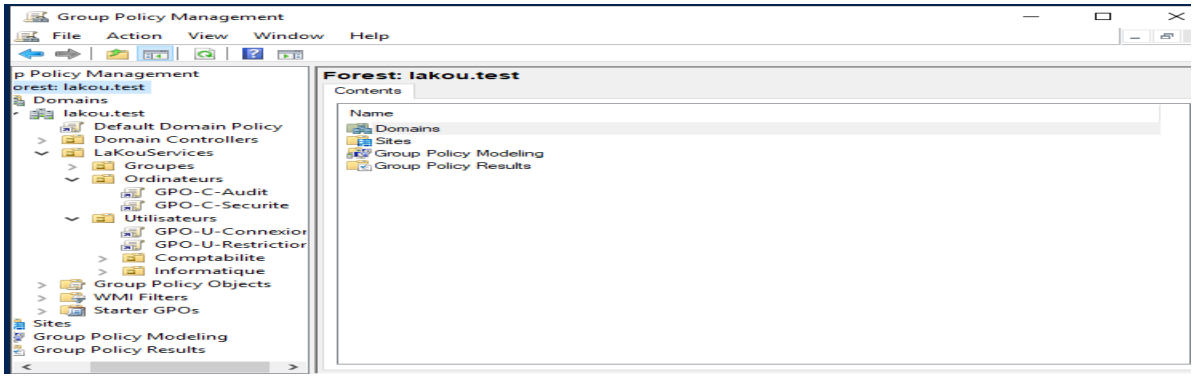




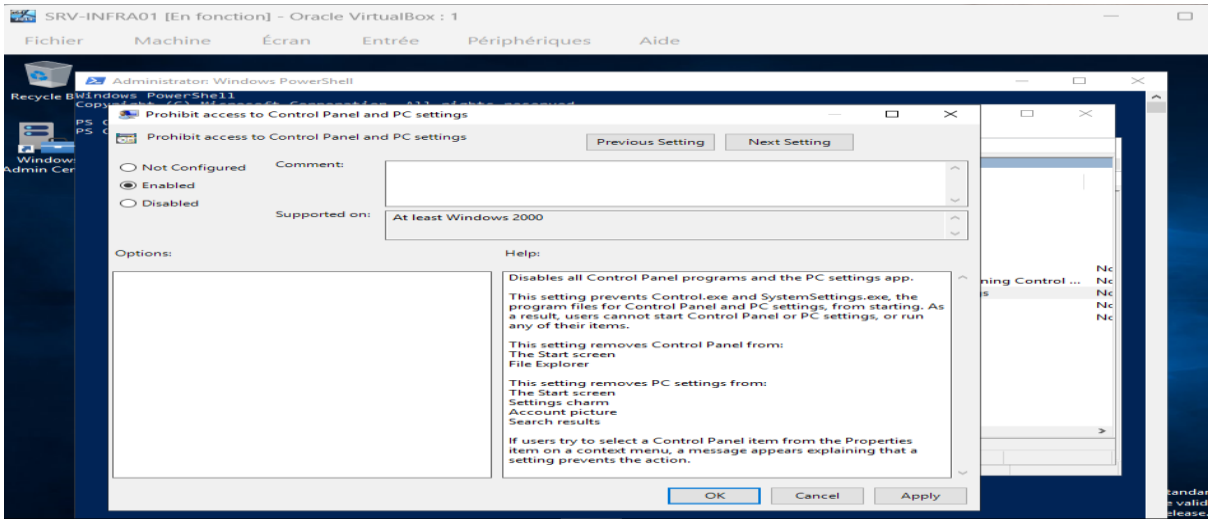
B. Créer les stratégies



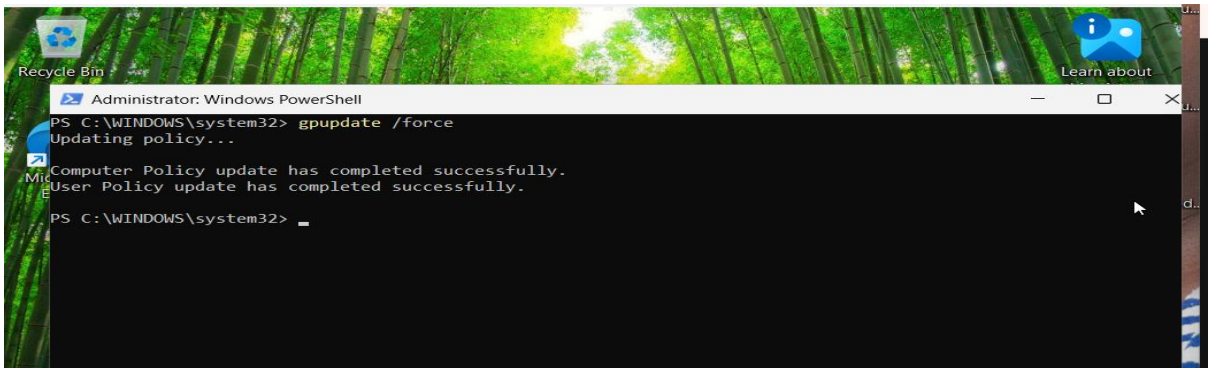


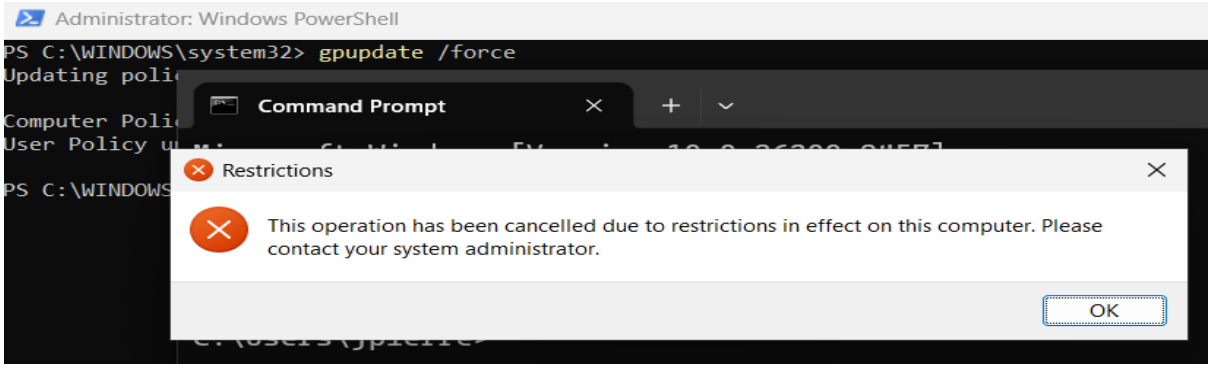


Activité 2: Restreindre certains paramètres utilisateur



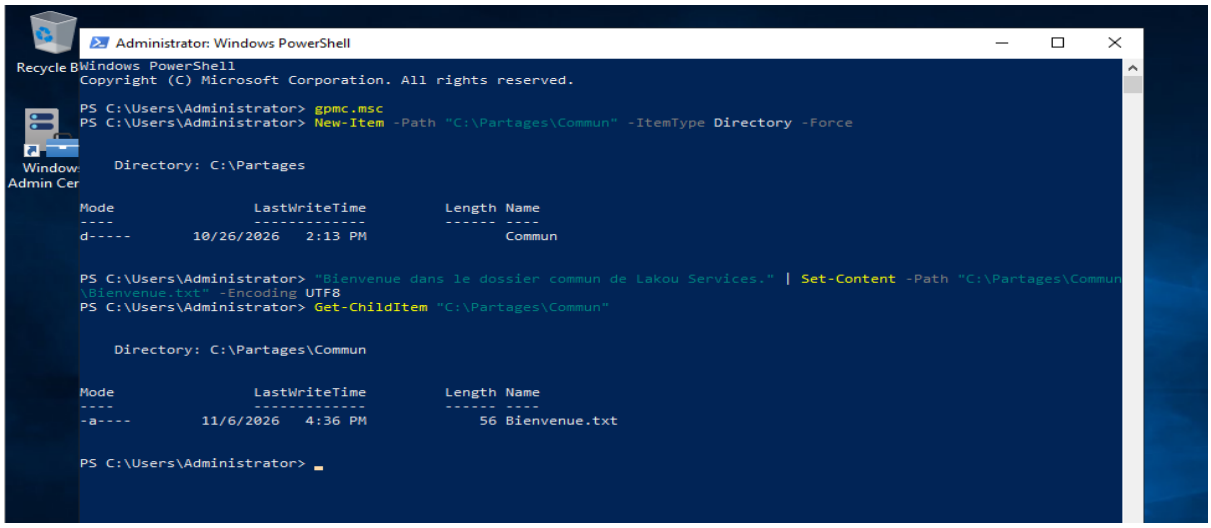
Tester sur CLIENT01



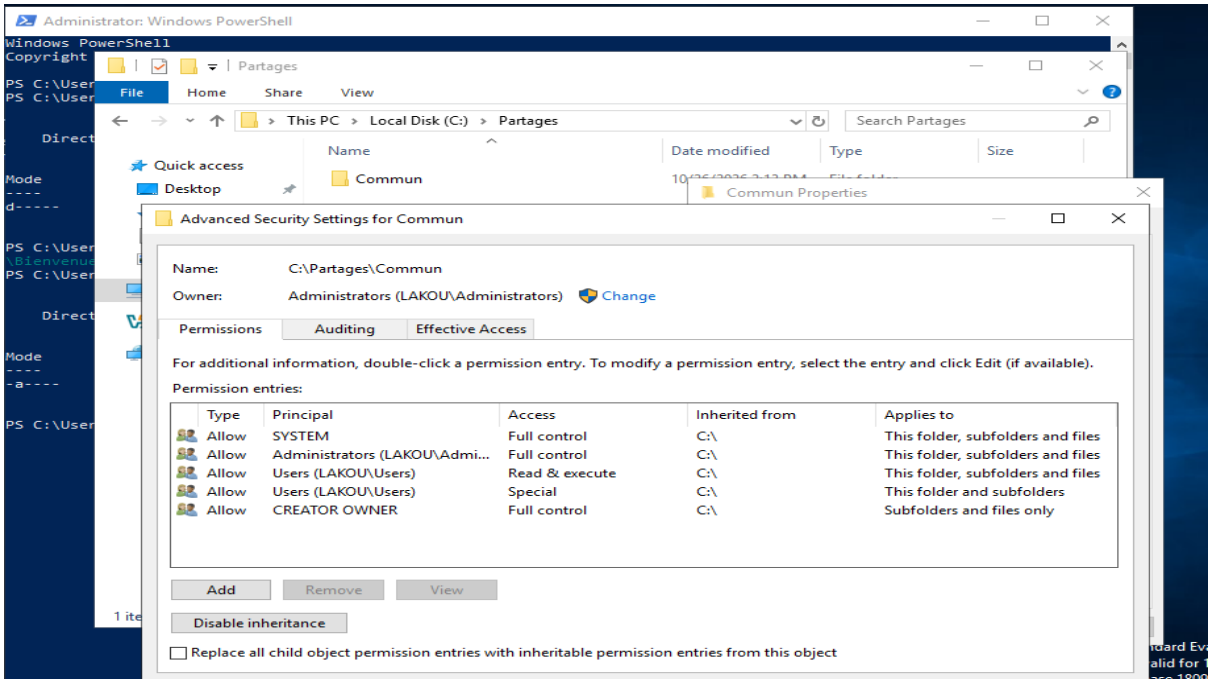


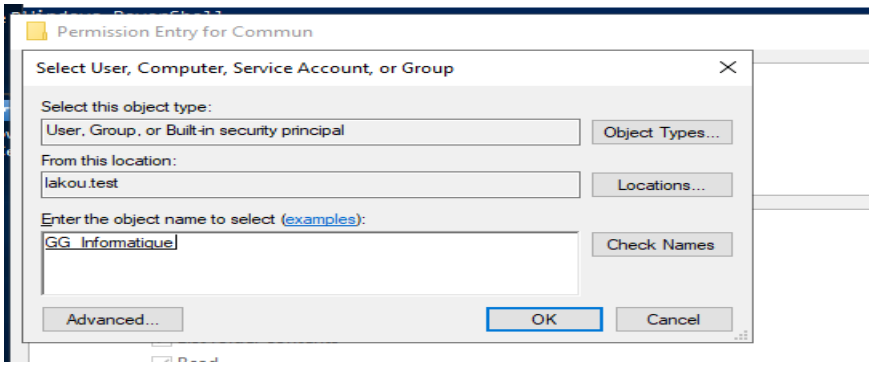
Activité 3: Préparer un dossier commun

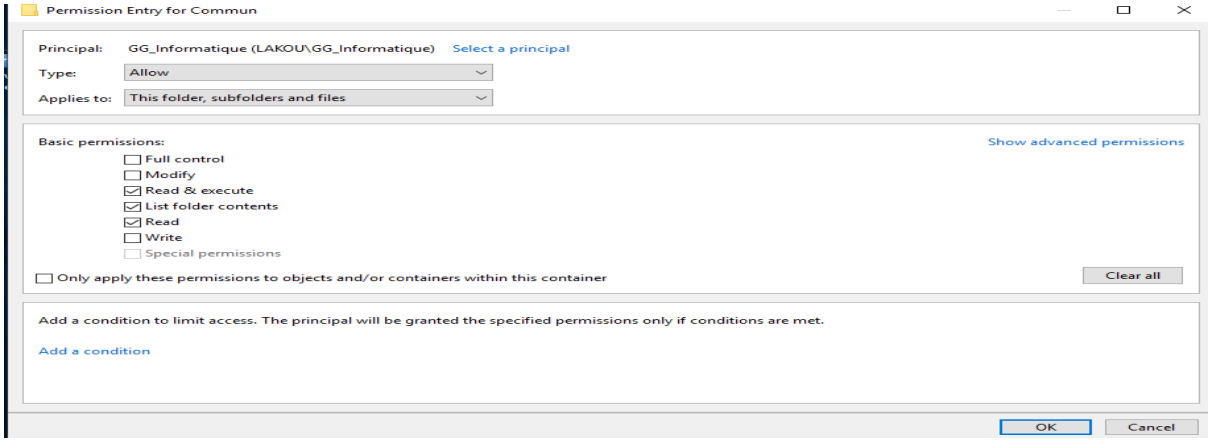
A. Créer le dossier

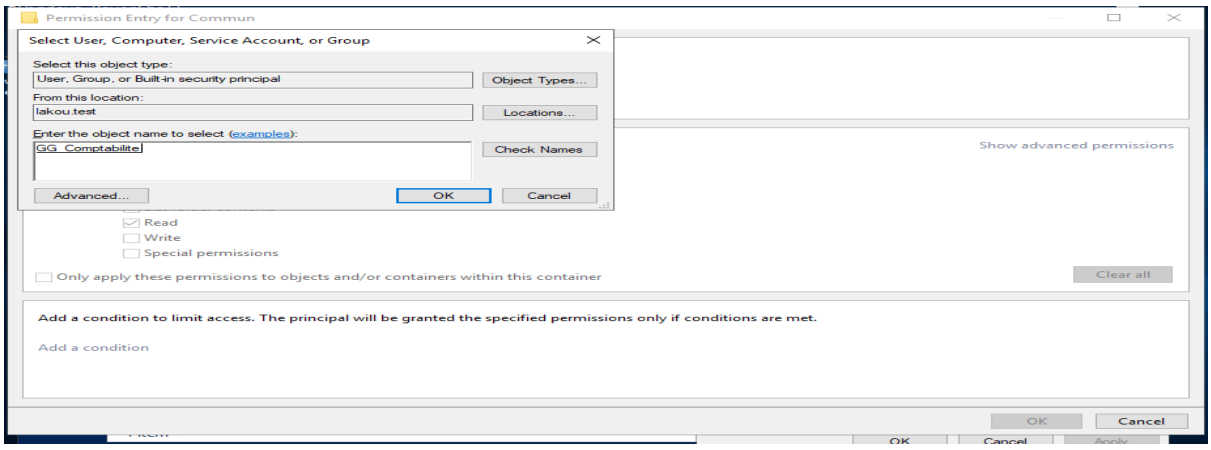


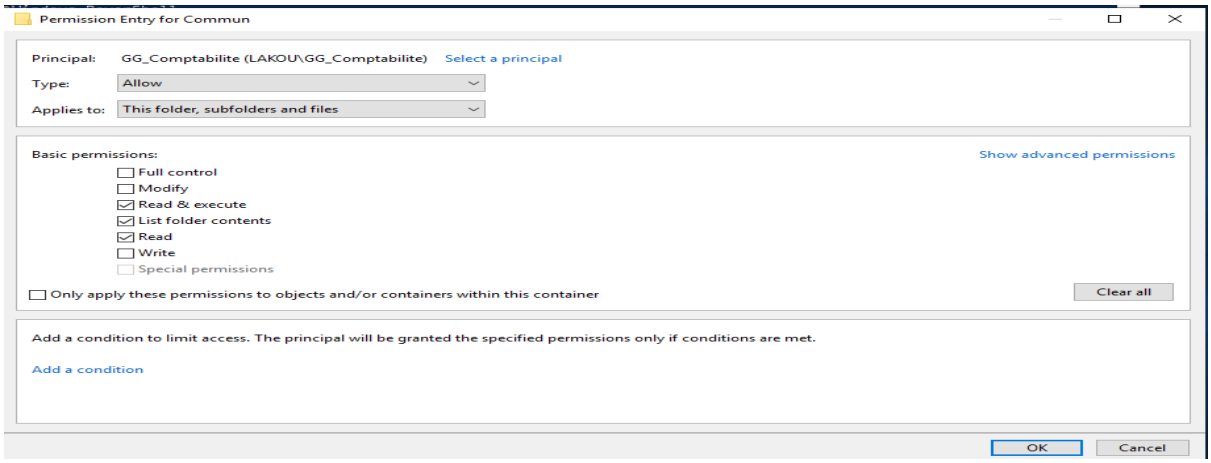
B. Configurer les autorisations NTFS

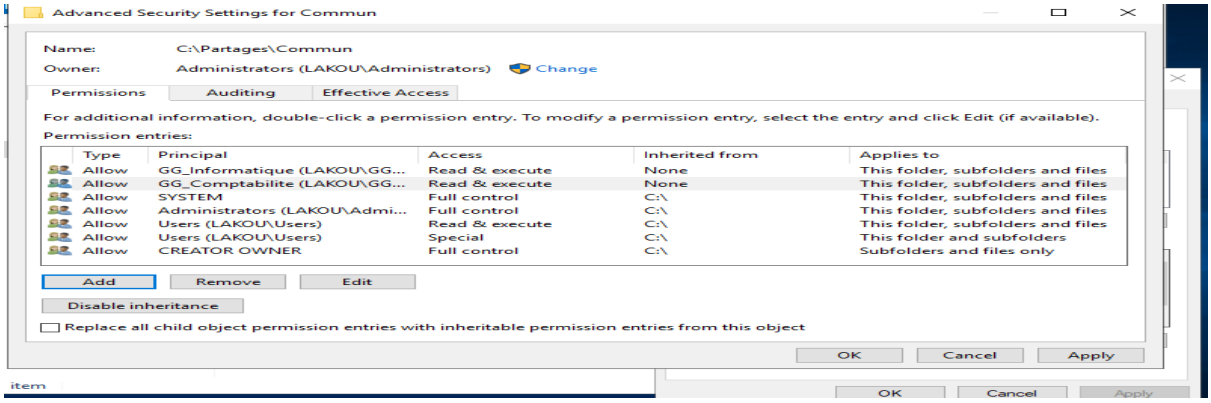



Permission Entry for Commun
Select User, Computer, Service Account, or Group
Select this object type:
User, Group, or Built-in security principal
Object Types...
From this location:
lakou.test
Locations...
Enter the object name to select (examples):
GG_Informatique
Check Names
Advanced...
OK
Cancel


Permission Entry for Commun
Principal: GG_Informatique (LAKOU\GG_Informatique) Select a principal
Type: Allow
Applies to: This folder, subfolders and files
Basic permissions:
Show advanced permissions
Full control
Modify
Read & execute
List folder contents
Read
Write
Special permissions
Only apply these permissions to objects and/or containers within this container
Clear all
Add a condition to limit access. The principal will be granted the specified permissions only if conditions are met.
Add a condition
OK
Cancel


Permission Entry for Commun
Select User, Computer, Service Account, or Group
Select this object type:
User, Group, or Built-in security principal
Object Types...
From this location:
lakou.test
Locations...
Enter the object name to select (examples):
GG_Comptabilite
Check Names
Advanced...
OK
Cancel
Show advanced permissions
Read
Write
Special permissions
Only apply these permissions to objects and/or containers within this container
Clear all
Add a condition to limit access. The principal will be granted the specified permissions only if conditions are met.
Add a condition
OK
Cancel


Permission Entry for Commun
Principal: GG_Comptabilite (LAKOU\GG_Comptabilite) Select a principal
Type: Allow
Applies to: This folder, subfolders and files
Basic permissions:
Show advanced permissions
Full control
Modify
Read & execute
List folder contents
Read
Write
Special permissions
Only apply these permissions to objects and/or containers within this container
Clear all
Add a condition to limit access. The principal will be granted the specified permissions only if conditions are met.
Add a condition
OK
Cancel



C. Créer le partage SMB

```
PS C:\Users\Administrator> New-SmbShare -Name "Commun" -Path "C:\Partages\Commun" -ReadAccess "LAKOU\GG_Informatique","LAKOU\GG_Comptabilite"

Name    ScopeName Path                 Description
----    --------- ----                 -----------
Commun  *         C:\Partages\Commun

PS C:\Users\Administrator> Get-SmbShare -Name "Commun"

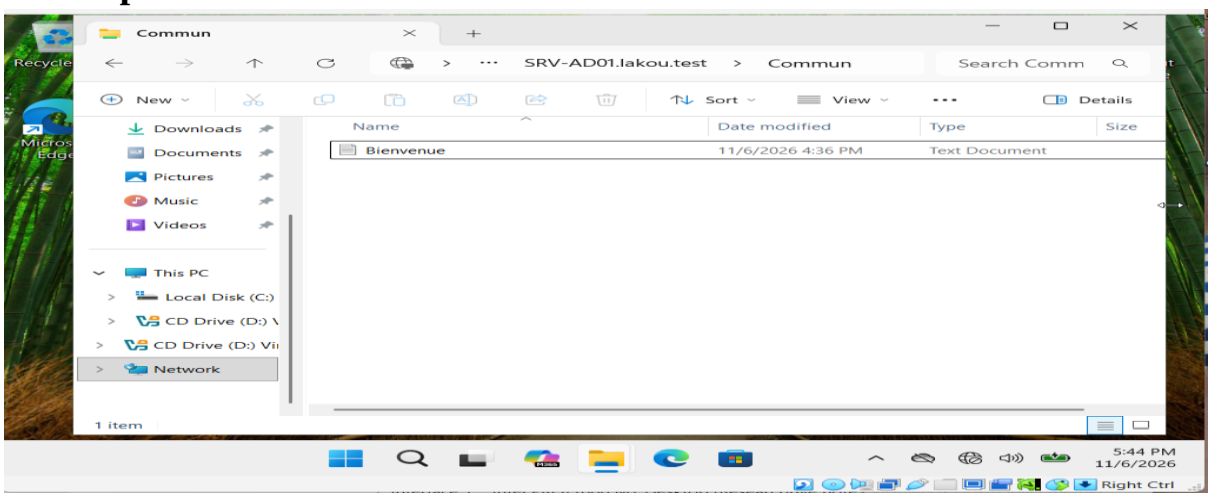
Name    ScopeName Path                 Description
----    --------- ----                 -----------
Commun  *         C:\Partages\Commun

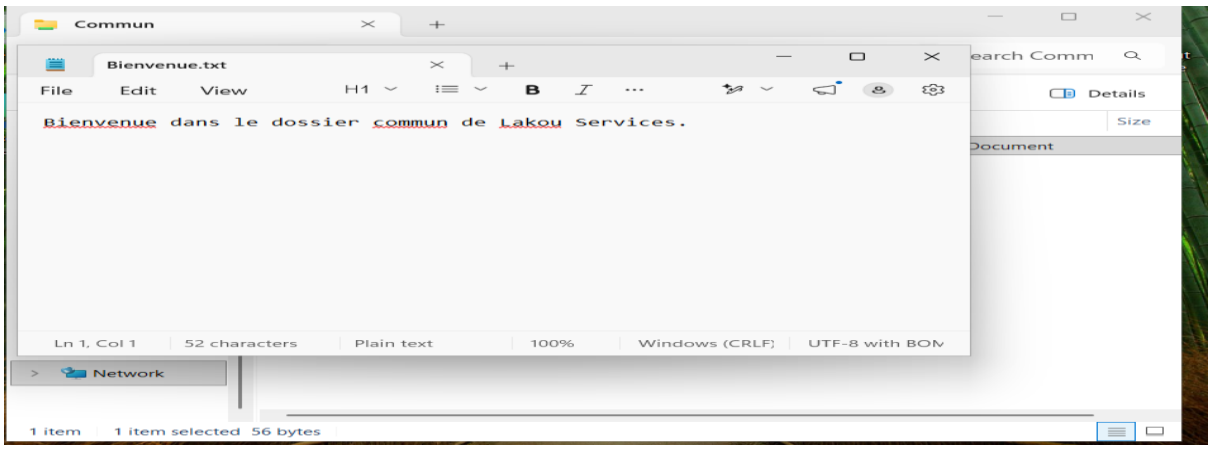
PS C:\Users\Administrator> Get-SmbShareAccess -Name "Commun"

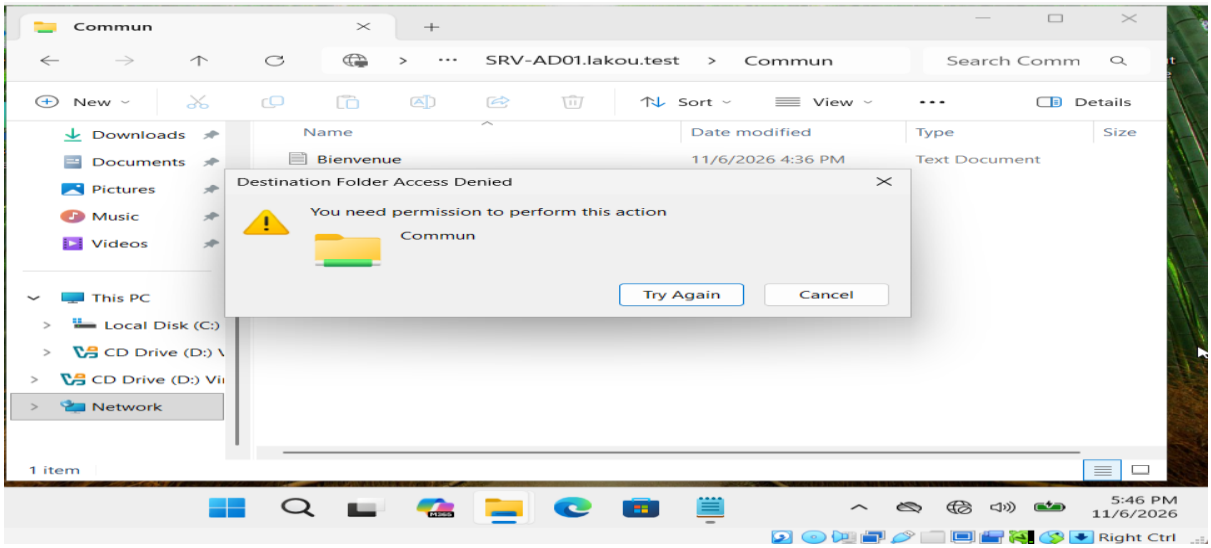
Name    ScopeName AccountName            AccessControlType AccessRight
----    --------- -----------            ----------------- -----------
Commun  *         LAKOU\GG_Informatique Allow             Read
Commun  *         LAKOU\GG_Comptabilite Allow             Read

PS C:\Users\Administrator>
```

Test depuis CLIENT01

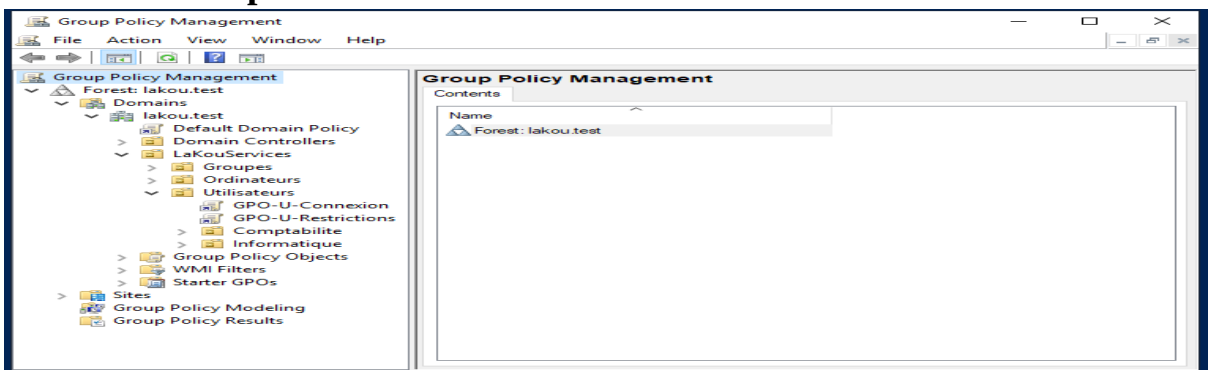


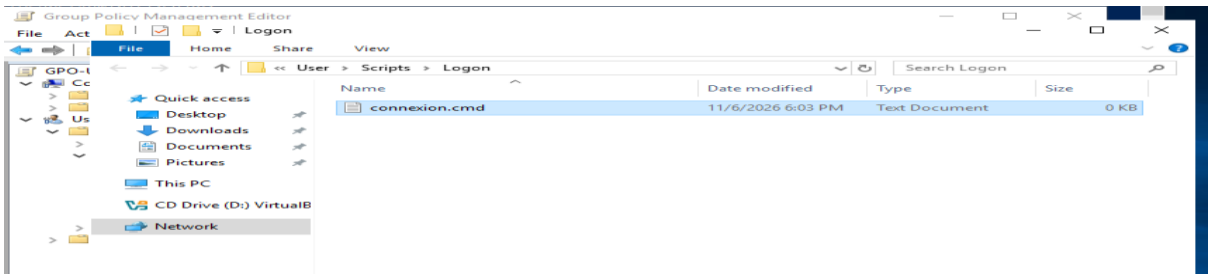


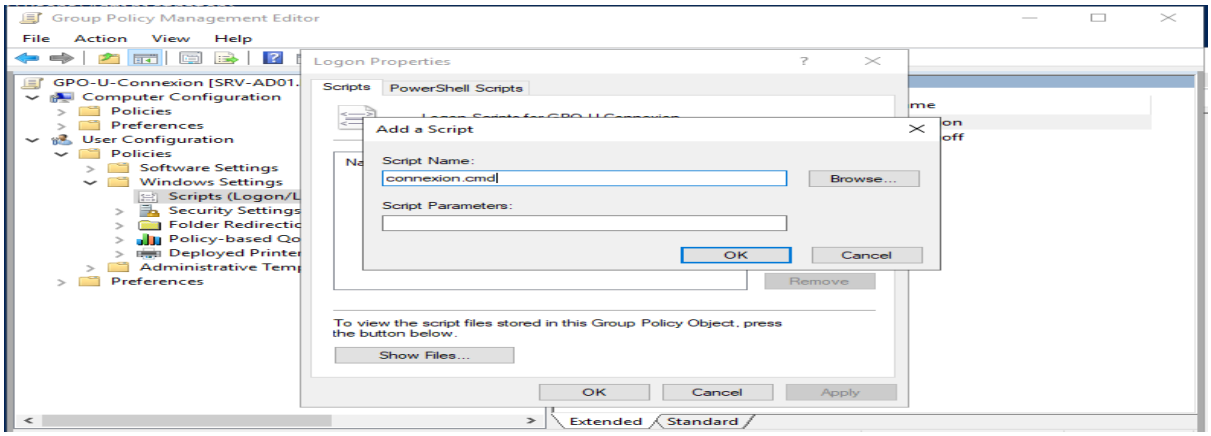


Activité 4: Déployer un script de connexion

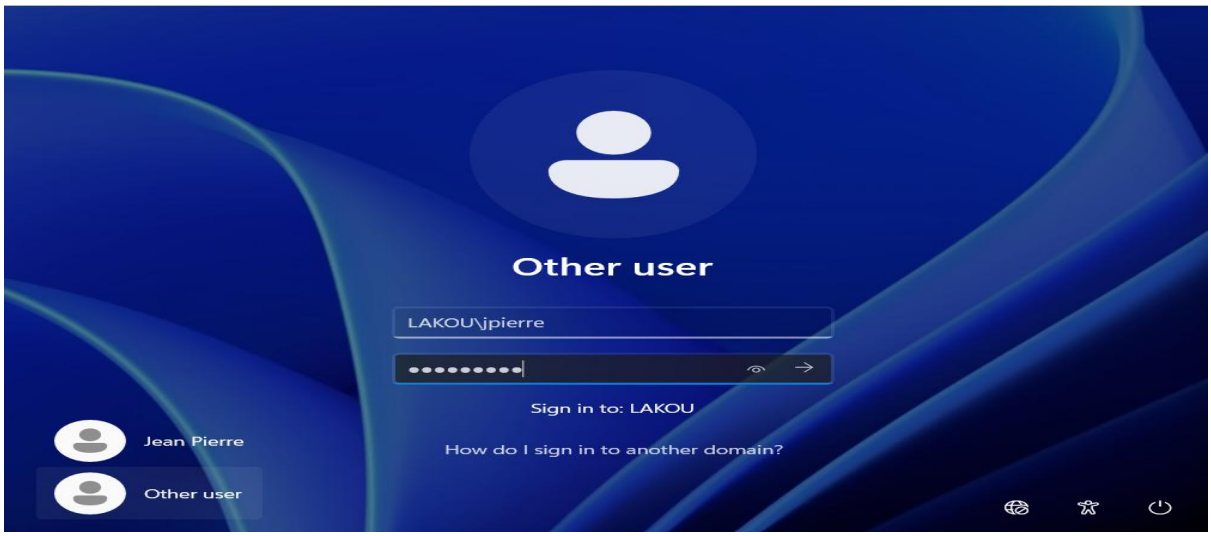
A. Créer le script

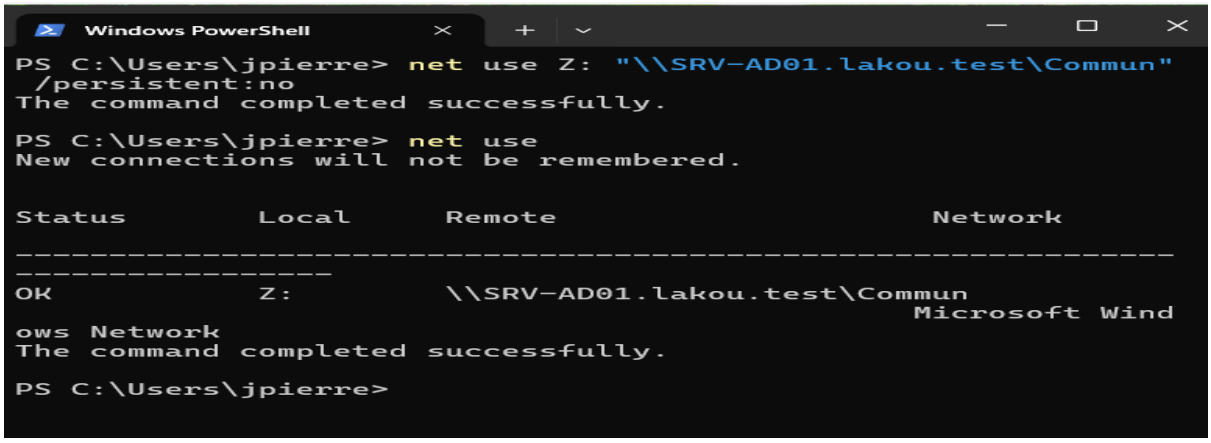


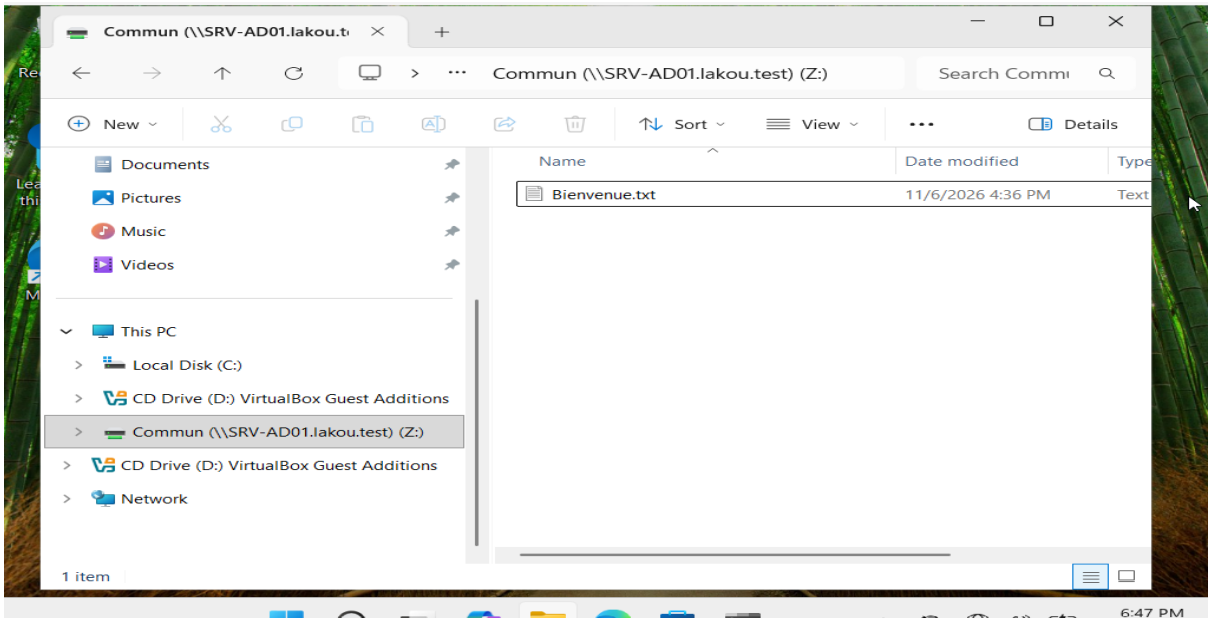


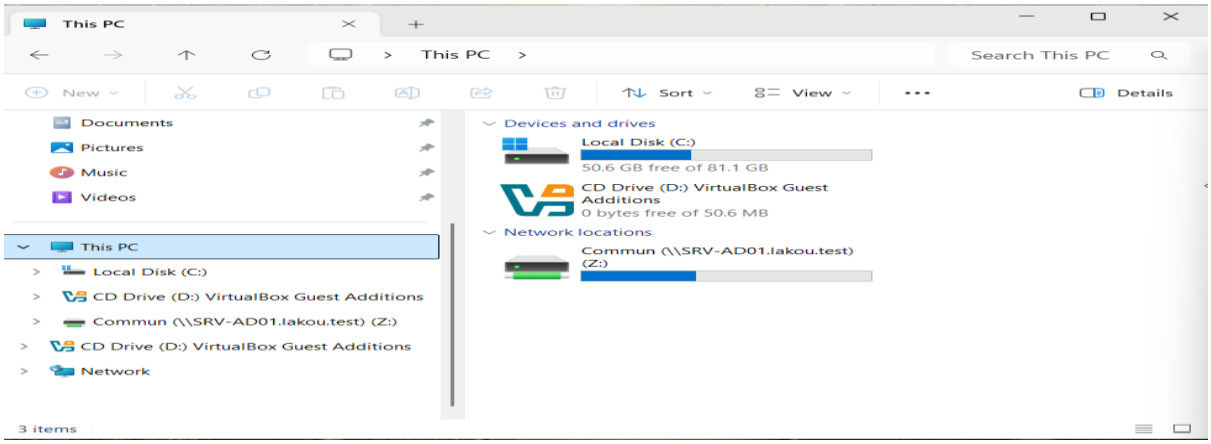


TESTER SUR CLIENT01

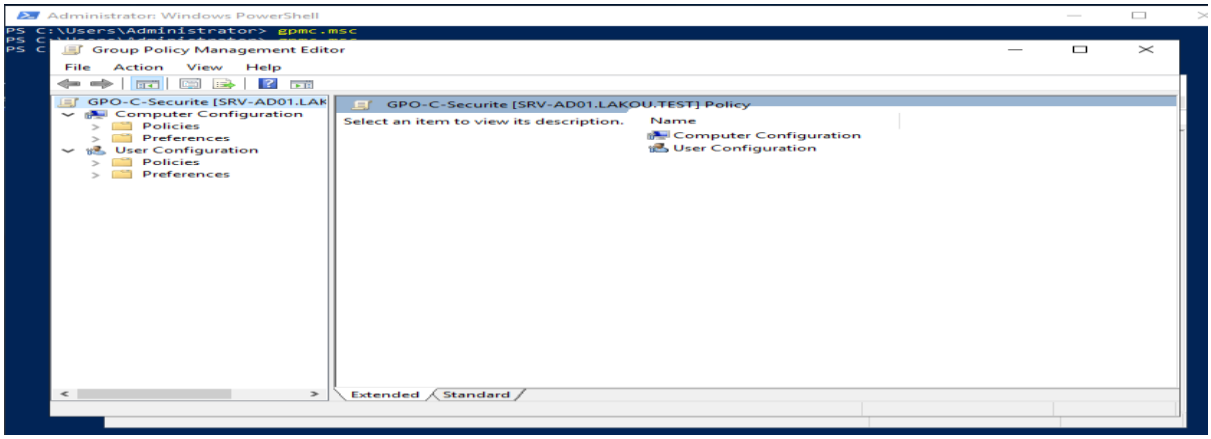


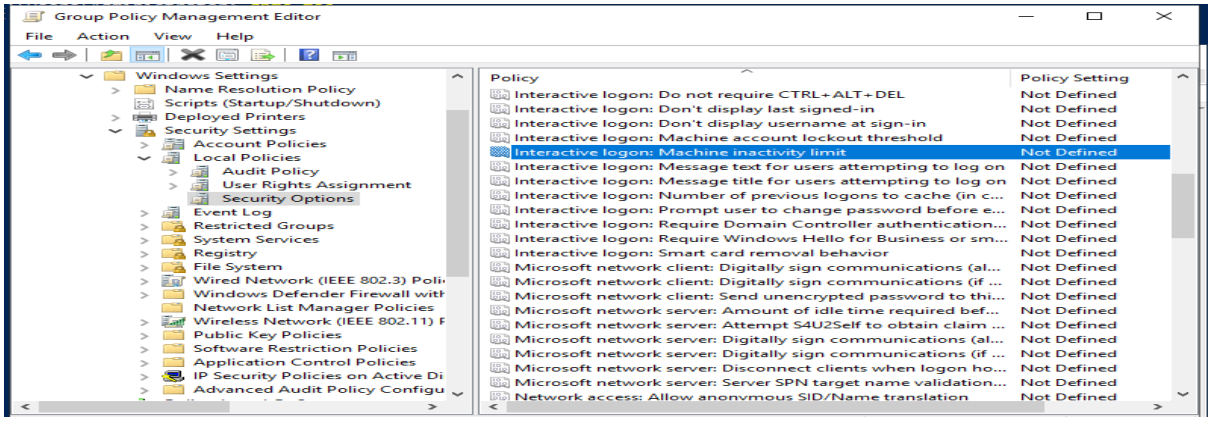


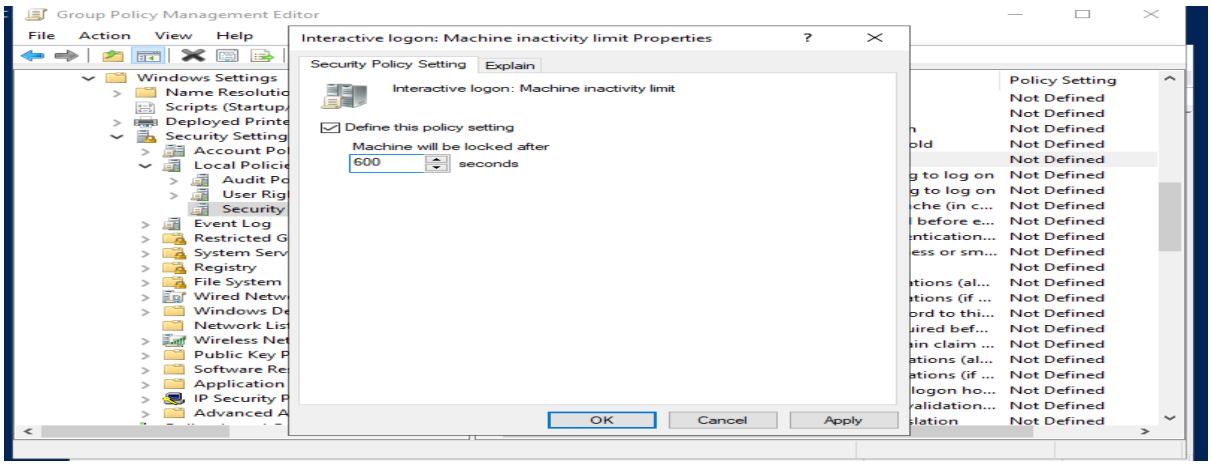


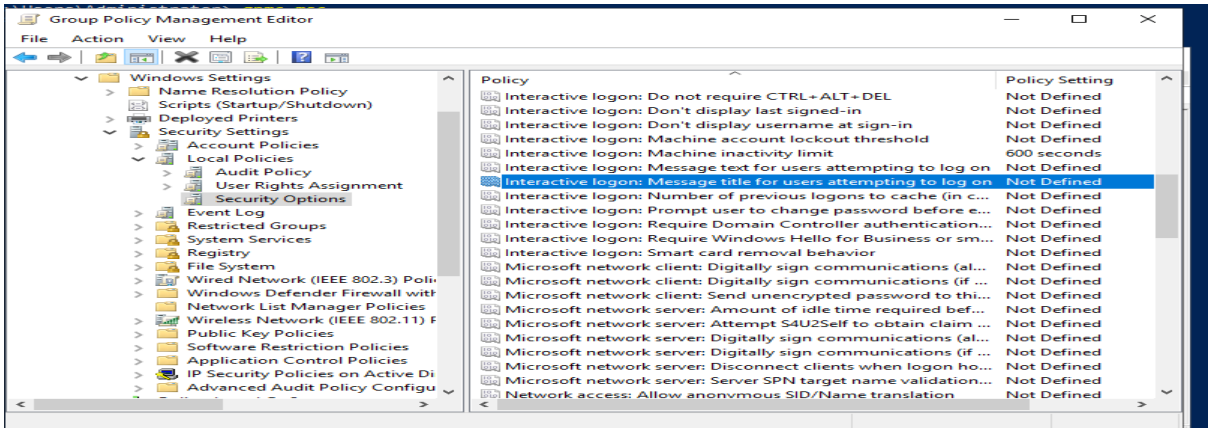


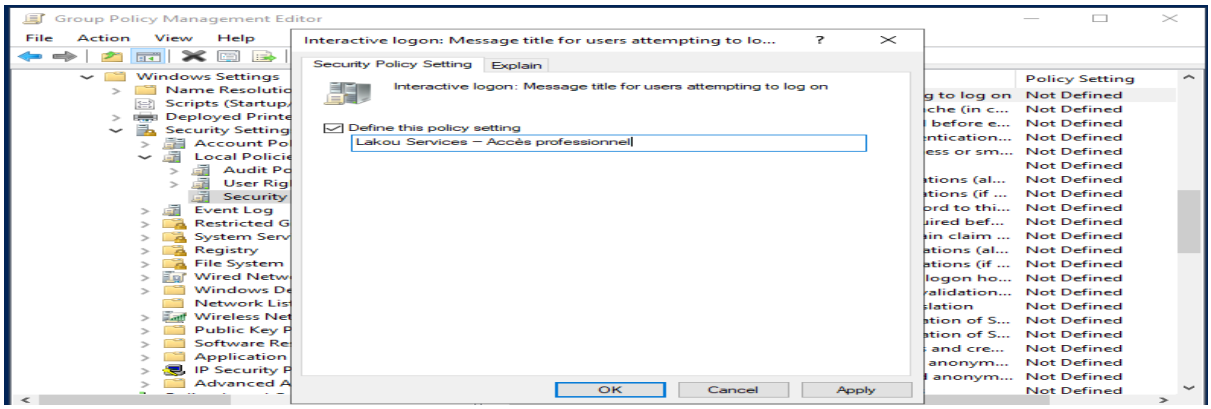
Activité 5: Renforcer la sécurité des postes

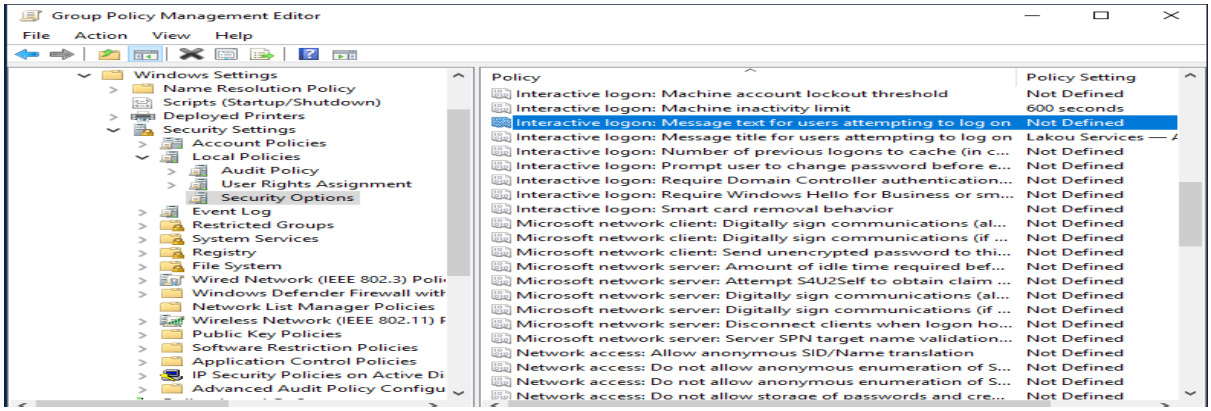


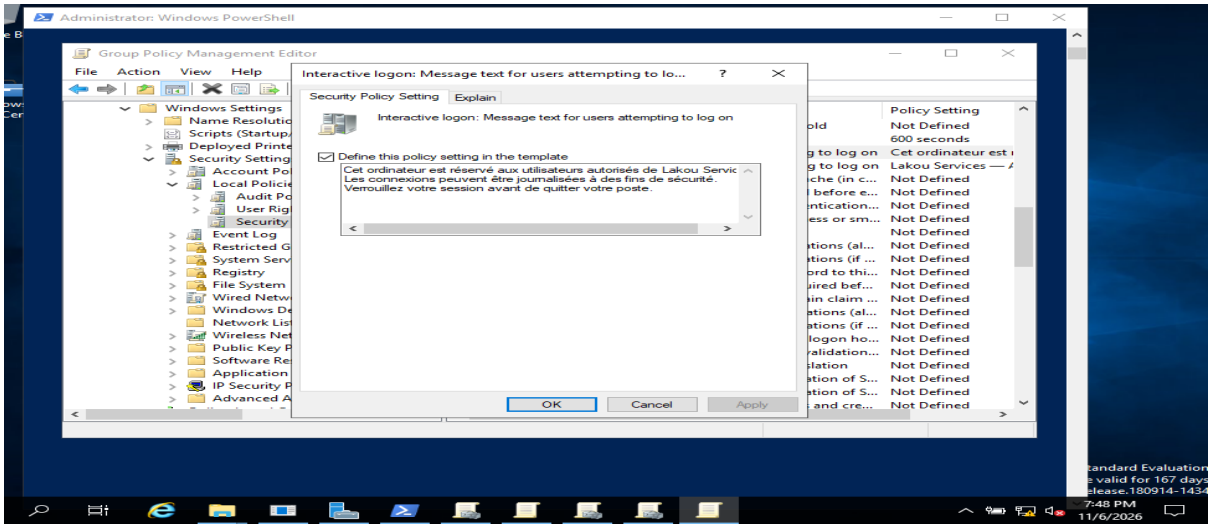


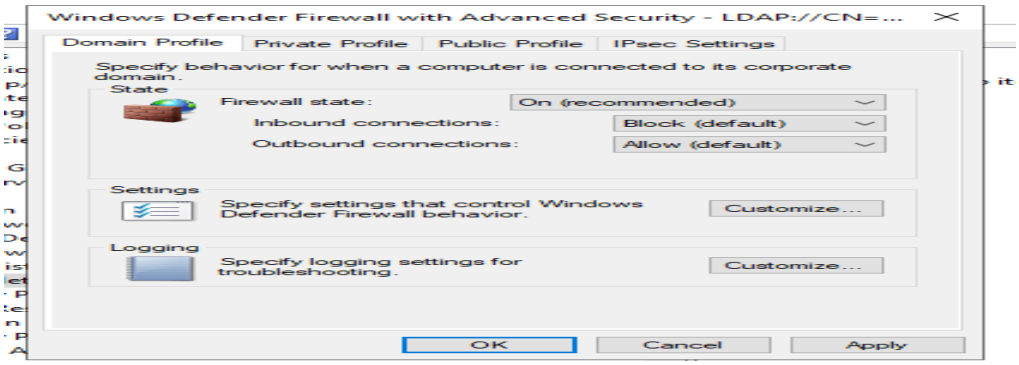


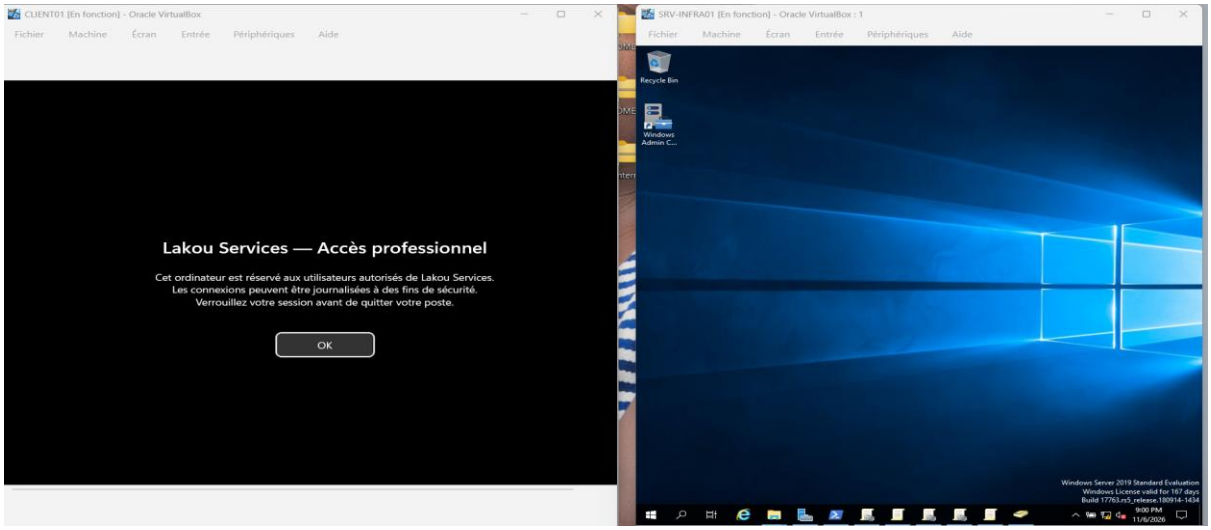

Group Policy Management Editor
File Action View Help
Windows Settings
Name Resolution Policy
Scripts (Startup/Shutdown)
Deployed Printers
Security Settings
Account Policies
Local Policies
Audit Policy
User Rights Assignment
Security Options
Event Log
Restricted Groups
System Services
Registry
File System
Wired Network (IEEE 802.3) Poli
Windows Defender Firewall with
Network List Manager Policies
Wireless Network (IEEE 802.11) F
Public Key Policies
Software Restriction Policies
Application Control Policies
IP Security Policies on Active Di
Advanced Audit Policy Configu
Policy
Policy Setting
Interactive logon: Do not require CTRL+ALT+DEL Not Defined
Interactive logon: Don't display last signed-in Not Defined
Interactive logon: Don't display username at sign-in Not Defined
Interactive logon: Machine account lockout threshold Not Defined
Interactive logon: Machine inactivity limit 600 seconds
Interactive logon: Message text for users attempting to log on Not Defined
Interactive logon: Message title for users attempting to log on Not Defined
Interactive logon: Number of previous logons to cache (in c... Not Defined
Interactive logon: Prompt user to change password before e... Not Defined
Interactive logon: Require Domain Controller authentication... Not Defined
Interactive logon: Require Windows Hello for Business or sm... Not Defined
Interactive logon: Smart card removal behavior Not Defined
Microsoft network client: Digitally sign communications (al... Not Defined
Microsoft network client: Digitally sign communications (if ... Not Defined
Microsoft network client: Send unencrypted password to thi... Not Defined
Microsoft network server: Amount of idle time required bef... Not Defined
Microsoft network server: Attempt S4U2Self to obtain claim ... Not Defined
Microsoft network server: Digitally sign communications (al... Not Defined
Microsoft network server: Digitally sign communications (if ... Not Defined
Microsoft network server: Disconnect clients when logon ho... Not Defined
Microsoft network server: Server SPN target name validatio... Not Defined
Network access: Allow anonymous SID/Name translation Not Defined

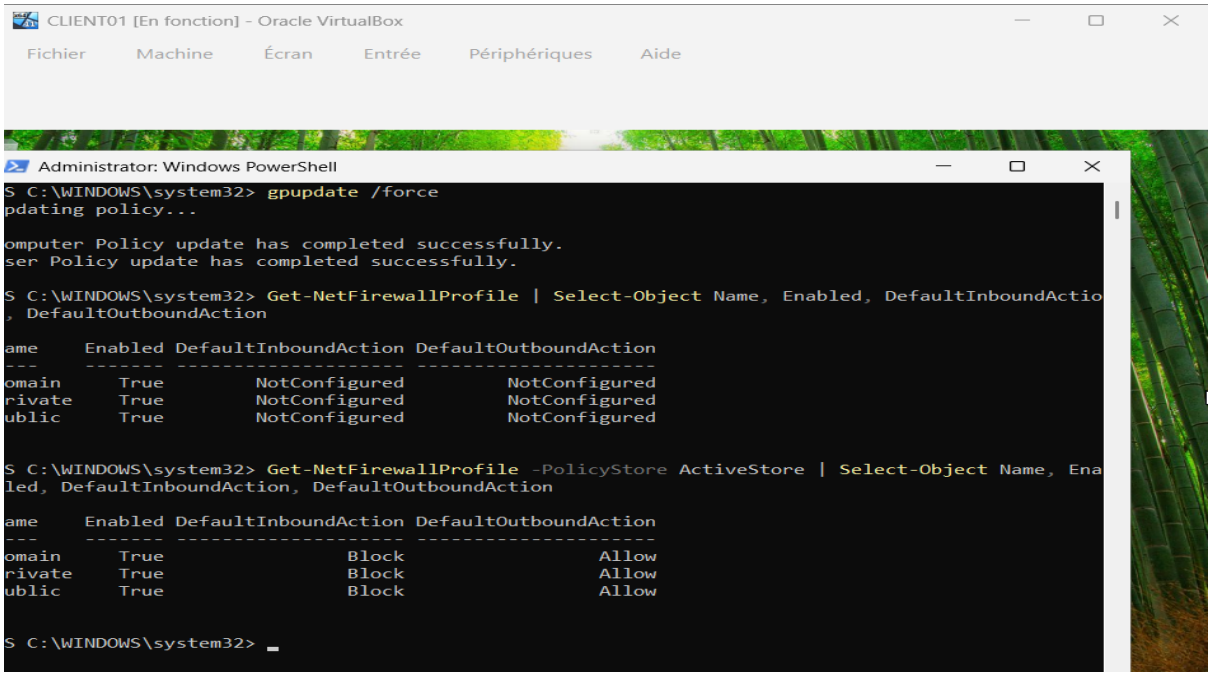

Group Policy Management Editor
File Action View Help
Interactive logon: Message title for users attempting to lo...
Security Policy Setting
Explain
Interactive logon: Message title for users attempting to log on
Define this policy setting
Lakou Services – Accès professionnel
OK
Cancel
Apply
Policy Setting


Group Policy Management Editor
File Action View Help
Windows Settings
Name Resolution Policy
Scripts (Startup/Shutdown)
Deployed Printers
Security Settings
Account Policies
Local Policies
Audit Policy
User Rights Assignment
Security Options
Event Log
Restricted Groups
System Services
Registry
File System
Wired Network (IEEE 802.3) Poli
Windows Defender Firewall with
Network List Manager Policies
Wireless Network (IEEE 802.11) F
Public Key Policies
Software Restriction Policies
Application Control Policies
IP Security Policies on Active Di
Advanced Audit Policy Configu
Policy
Policy Setting
Interactive logon: Machine account lockout threshold Not Defined
Interactive logon: Machine inactivity limit 600 seconds
Interactive logon: Message text for users attempting to log on Not Defined
Interactive logon: Message title for users attempting to log on Lakou Services —
Interactive logon: Number of previous logons to cache (in c... Not Defined
Interactive logon: Prompt user to change password before e... Not Defined
Interactive logon: Require Domain Controller authentication... Not Defined
Interactive logon: Require Windows Hello for Business or sm... Not Defined
Interactive logon: Smart card removal behavior Not Defined
Microsoft network client: Digitally sign communications (al... Not Defined
Microsoft network client: Digitally sign communications (if ... Not Defined
Microsoft network client: Send unencrypted password to thi... Not Defined
Microsoft network server: Amount of idle time required bef... Not Defined
Microsoft network server: Attempt S4U2Self to obtain claim ... Not Defined
Microsoft network server: Digitally sign communications (al... Not Defined
Microsoft network server: Digitally sign communications (if ... Not Defined
Microsoft network server: Disconnect clients when logon ho... Not Defined
Microsoft network server: Server SPN target name validation... Not Defined
Network access: Allow anonymous SID/Name translation Not Defined
Network access: Do not allow anonymous enumeration of S... Not Defined
Network access: Do not allow anonymous enumeration of S... Not Defined
Network access: Do not allow storage of passwords and cre... Not Defined

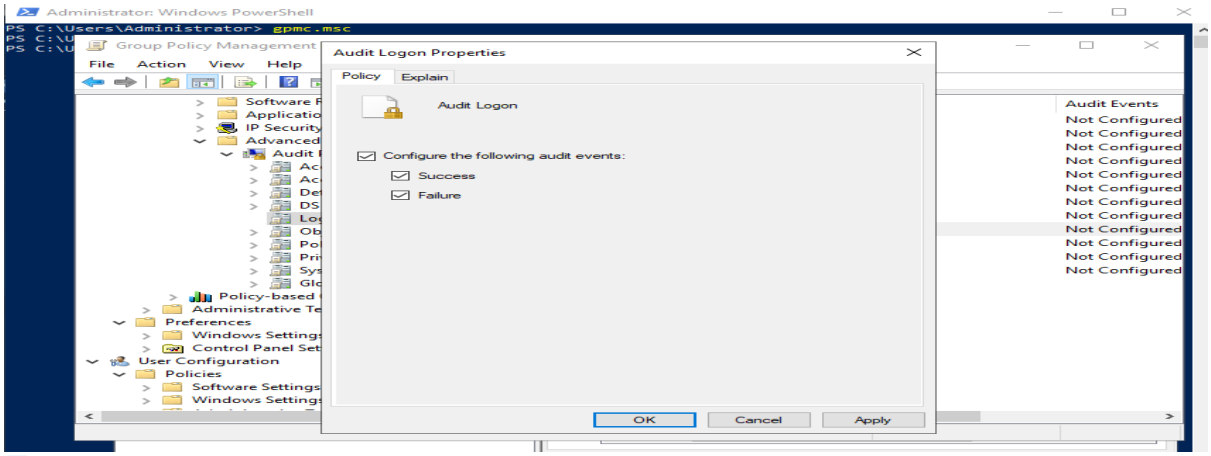

Administrator: Windows PowerShell
Group Policy Management Editor
File Action View Help
Interactive logon: Message text for users attempting to lo...
Security Policy Setting
Explain
Interactive logon: Message text for users attempting to log on
Define this policy setting in the template
Cet ordinateur est réservé aux utilisateurs autorisés de Lakou Servic
Les connexions peuvent être journalisées à des fins de sécurité.
Verrouillez votre session avant de quitter votre poste.
OK
Cancel
Apply
Policy Setting
Cet ordinateur est r
Lakou Services —
Standard Evaluation
valid for 167 days
release.180914-1434
7:48 PM
11/6/2026

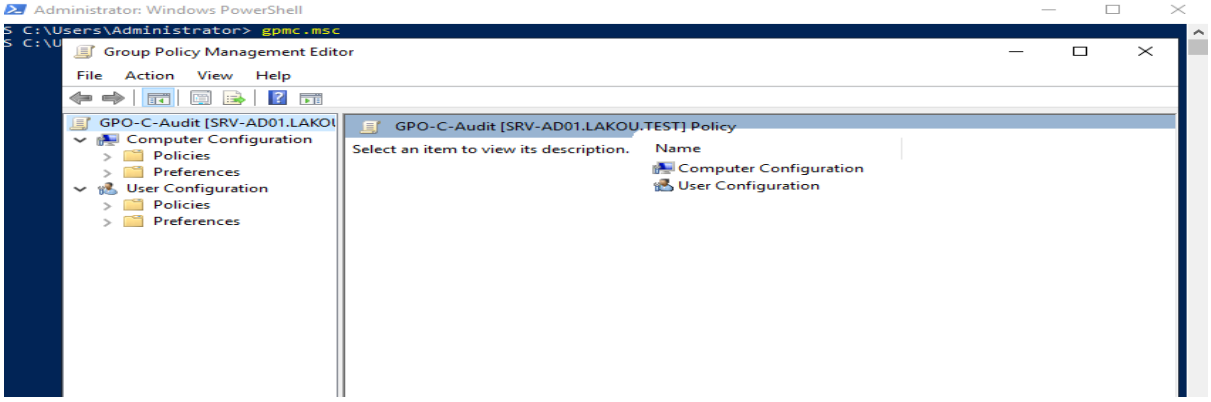


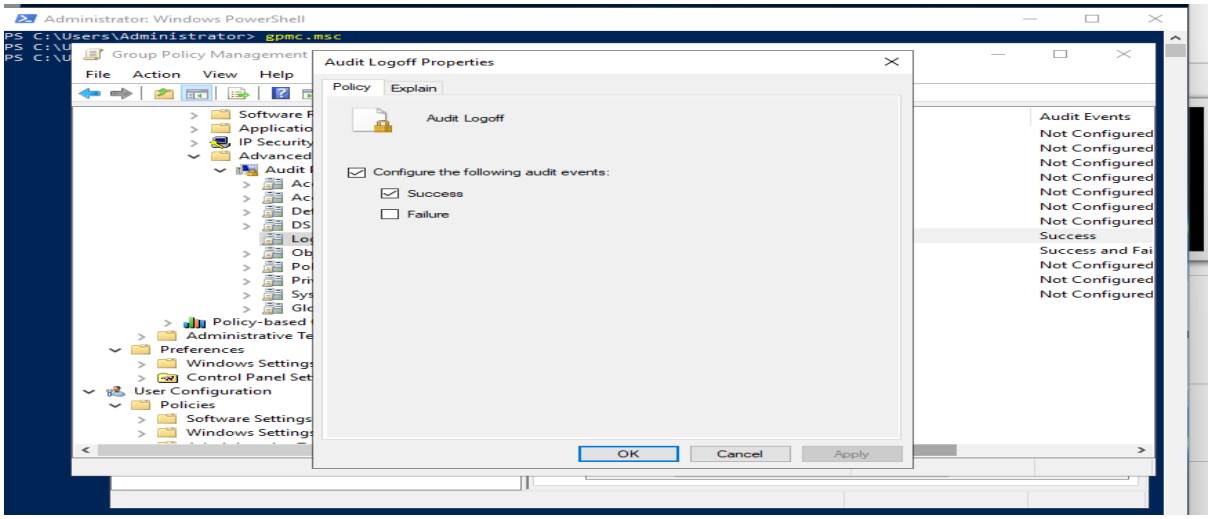


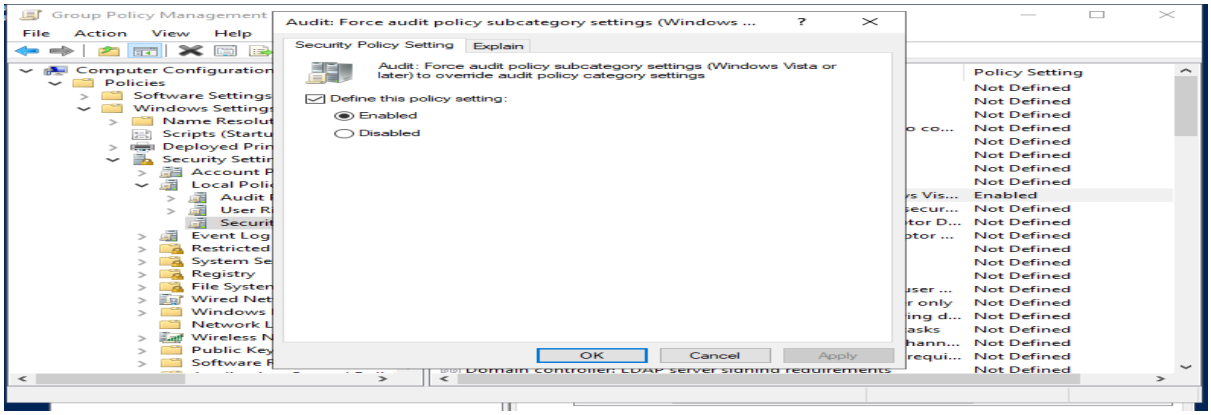


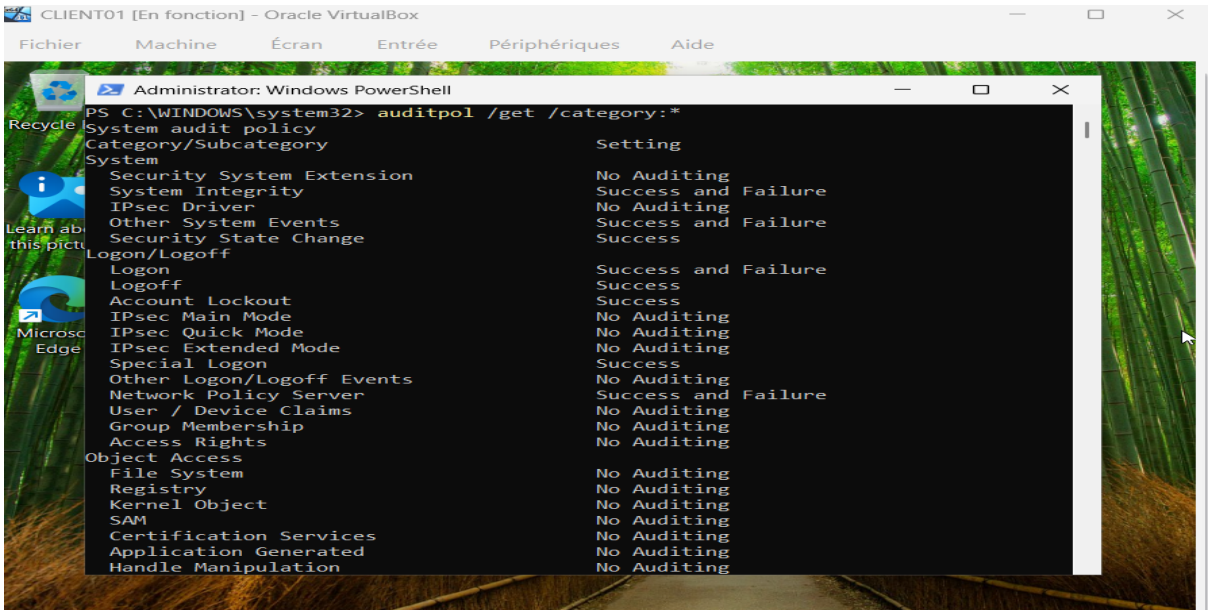
Activité 6: Activer l’audit des connexions

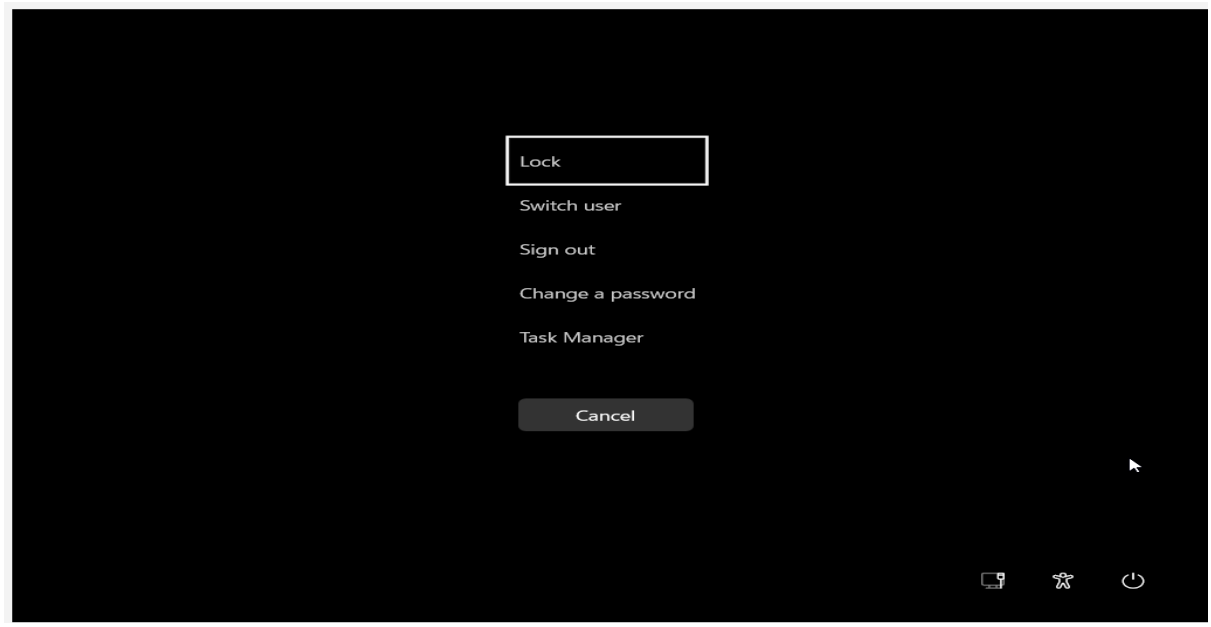

Administrator: Windows PowerShell
PS C:\Users\Administrator> gpmc.msc
Group Policy Management
File Action View Help
Audit Logon Properties
Policy Explain
Audit Logon
Configure the following audit events:
Success
Failure
OK Cancel Apply
Audit Events
Not Configured

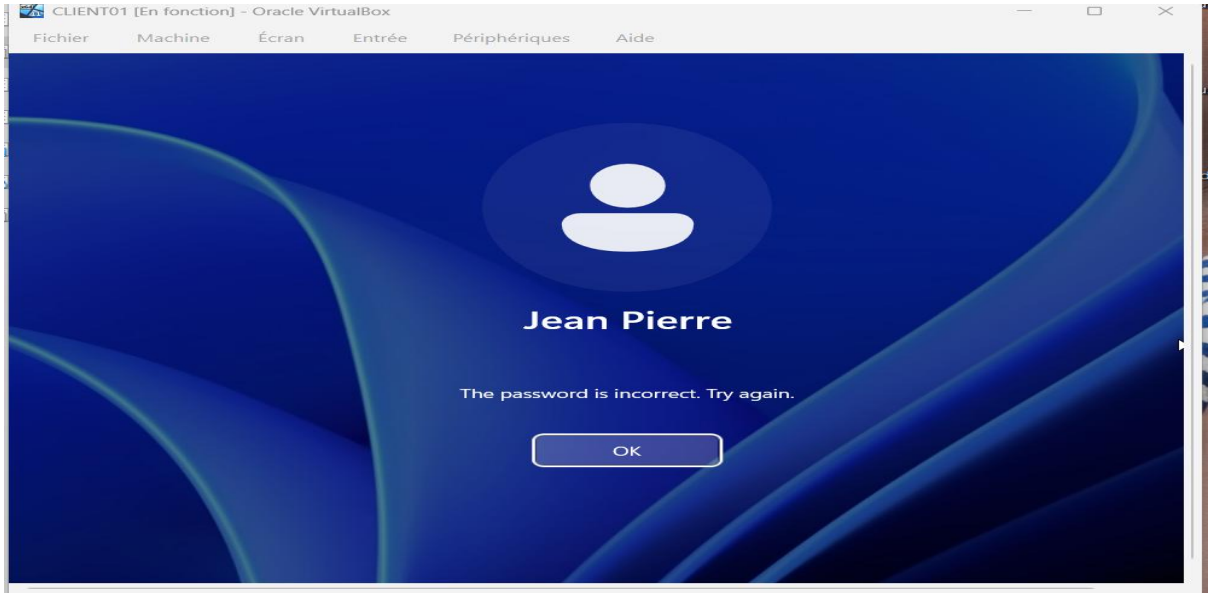

Administrator: Windows PowerShell
PS C:\Users\Administrator> gpmc.msc
Group Policy Management Editor
File Action View Help
GPO-C-Audit [SRV-AD01.LAKOU
Computer Configuration
Policies
Preferences
User Configuration
Policies
Preferences
GPO-C-Audit [SRV-AD01.LAKOU.TEST] Policy
Select an item to view its description.
Name
Computer Configuration
User Configuration

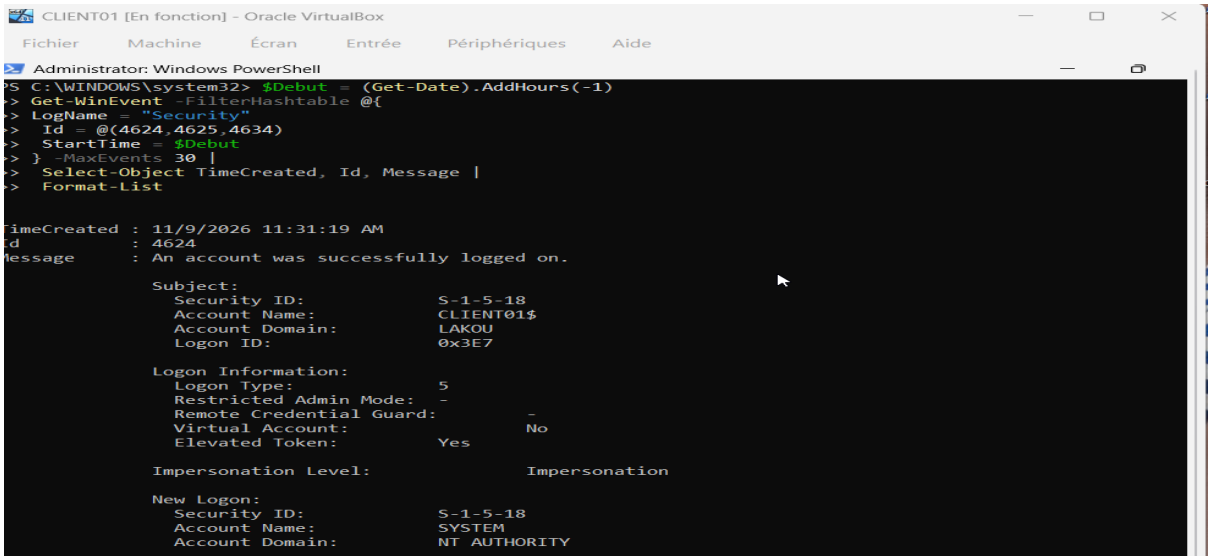

Administrator: Windows PowerShell
PS C:\Users\Administrator> gpmc.msc
Group Policy Management
File Action View Help
Audit Logoff Properties
Policy Explain
Audit Logoff
Configure the following audit events:
Success
Failure
OK Cancel Apply
Audit Events
Not Configured
Success
Success and Fai


Group Policy Management
File Action View Help
Computer Configuration
Policies
Software Settings
Windows Settings
Audit: Force audit policy subcategory settings (Windows ...
Security Policy Setting
Explain
Audit: Force audit policy subcategory settings (Windows Vista or later) to override audit policy category settings
Define this policy setting:
Enabled
Disabled
OK
Cancel
Apply
Policy Setting
Not Defined
Enabled


CLIENT01 [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> auditpol /get /category:*
System audit policy
Category/Subcategory Setting
System
Security System Extension No Auditing
System Integrity Success and Failure
IPsec Driver No Auditing
Other System Events Success and Failure
Security State Change Success
Logon/Logoff
Logon Success and Failure
Logoff Success
Account Lockout Success
IPsec Main Mode No Auditing
IPsec Quick Mode No Auditing
IPsec Extended Mode No Auditing
Special Logon Success
Other Logon/Logoff Events No Auditing
Network Policy Server Success and Failure
User / Device Claims No Auditing
Group Membership No Auditing
Access Rights No Auditing
Object Access
File System No Auditing
Registry No Auditing
Kernel Object No Auditing
SAM No Auditing
Certification Services No Auditing
Application Generated No Auditing
Handle Manipulation No Auditing
Recycle
Edge


Lock
Switch user
Sign out
Change a password
Task Manager
Cancel






```
PS C:\WINDOWS\system32> $Debut = (Get-Date).AddHours(-1)
> Get-WinEvent -FilterHashtable @{
>  LogName = "Security"
>    Id = @(4624,4625,4634)
>    StartTime = $Debut
> } -MaxEvents 30 |
>   Select-Object TimeCreated, Id, Message |
>   Format-List

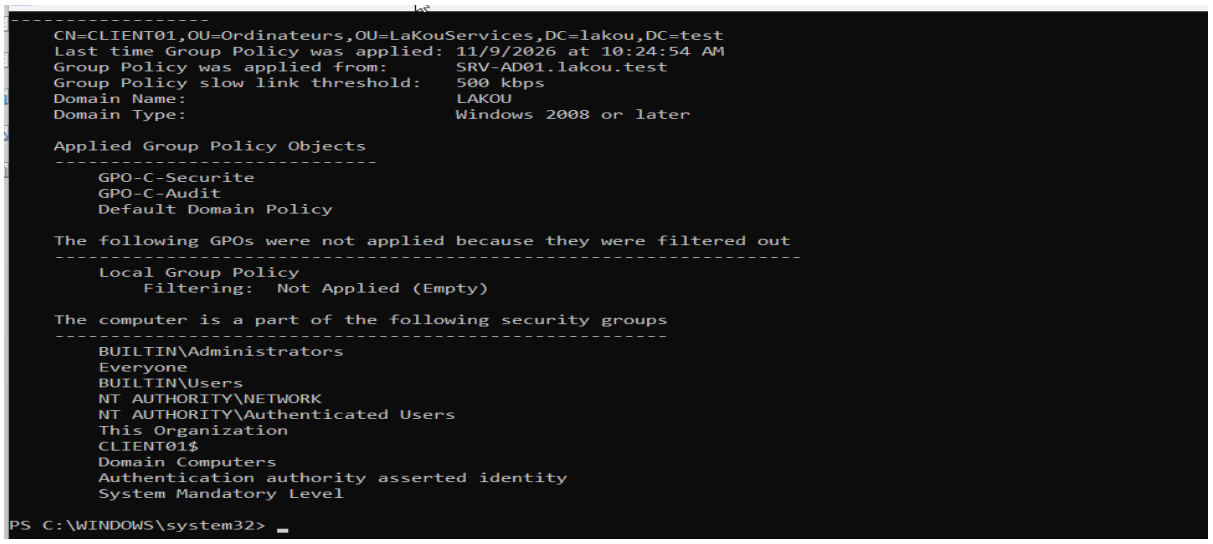
TimeCreated : 11/9/2026 11:31:19 AM
Id          : 4624
Message     : An account was successfully logged on.

              Subject:
                Security ID:            S-1-5-18
                Account Name:           CLIENT01$
                Account Domain:         LAKOU
                Logon ID:               0x3E7

              Logon Information:
                Logon Type:             5
                Restricted Admin Mode:  -
                Remote Credential Guard:        -
                Virtual Account:                No
                Elevated Token:         Yes

              Impersonation Level:              Impersonation

              New Logon:
                Security ID:            S-1-5-18
                Account Name:           SYSTEM
                Account Domain:         NT AUTHORITY
```



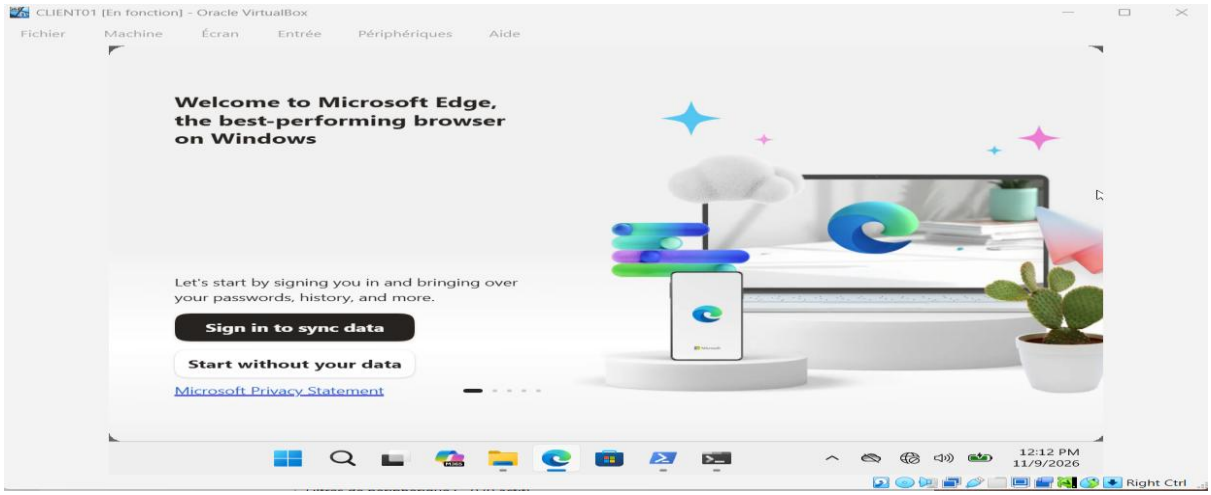
```
-------------------
    CN=CLIENT01,OU=Ordinateurs,OU=LaKouServices,DC=lakou,DC=test
    Last time Group Policy was applied: 11/9/2026 at 10:24:54 AM
    Group Policy was applied from:      SRV-AD01.lakou.test
    Group Policy slow link threshold:   500 kbps
    Domain Name:                        LAKOU
    Domain Type:                        Windows 2008 or later

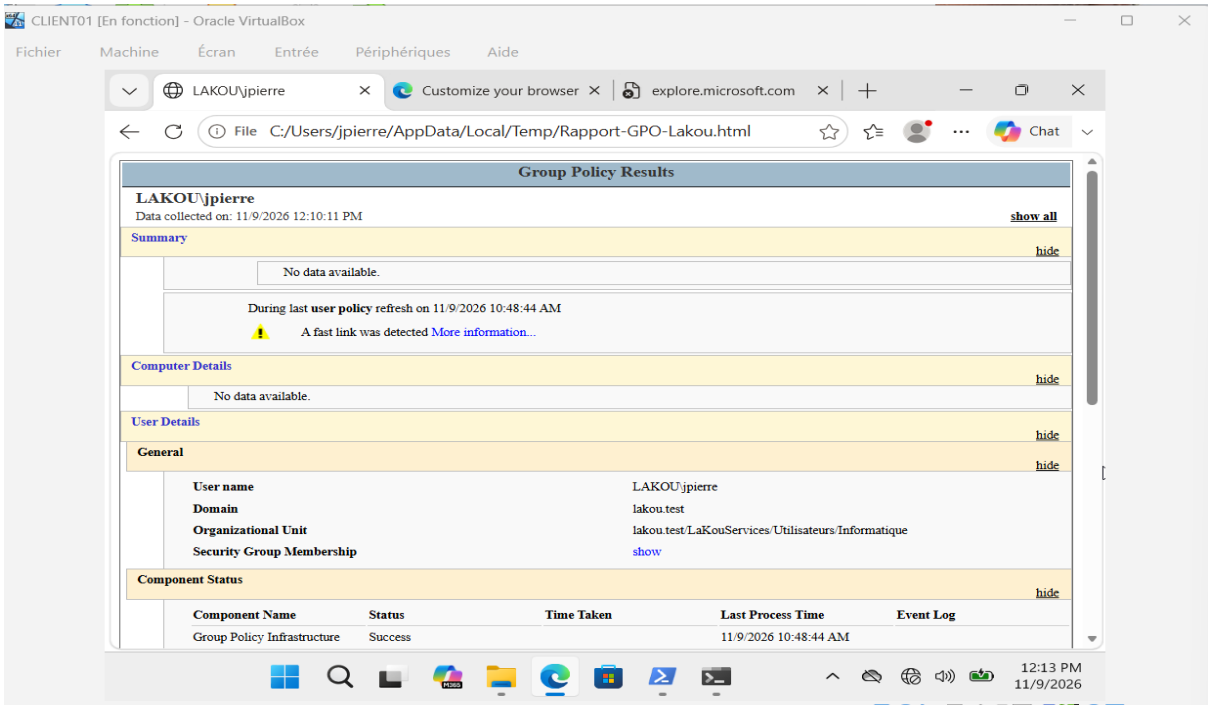
    Applied Group Policy Objects
    -----------------------------
        GPO-C-Securite
        GPO-C-Audit
        Default Domain Policy

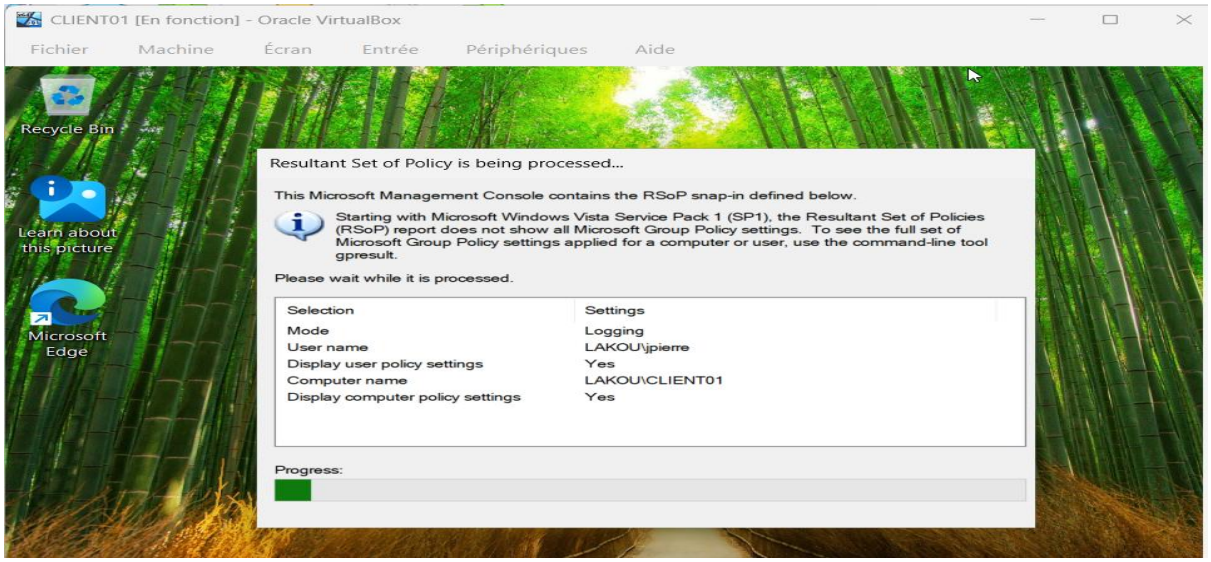
    The following GPOs were not applied because they were filtered out
    -------------------------------------------------------------------
        Local Group Policy
            Filtering:  Not Applied (Empty)

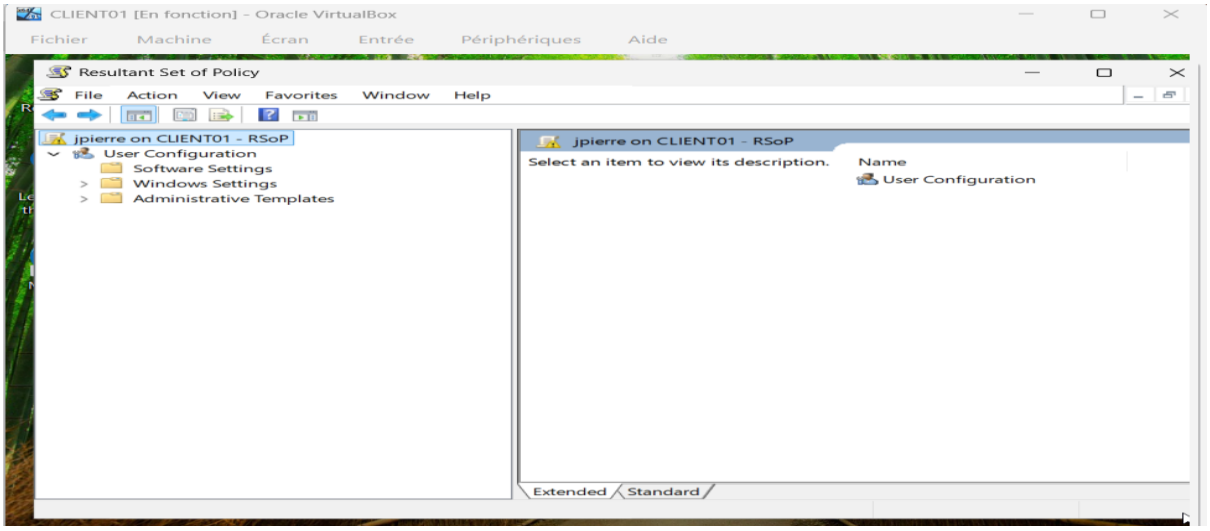
    The computer is a part of the following security groups
    -------------------------------------------------------
        BUILTIN\Administrators
        Everyone
        BUILTIN\Users
        NT AUTHORITY\NETWORK
        NT AUTHORITY\Authenticated Users
        This Organization
        CLIENT01$
        Domain Computers
        Authentication authority asserted identity
        System Mandatory Level

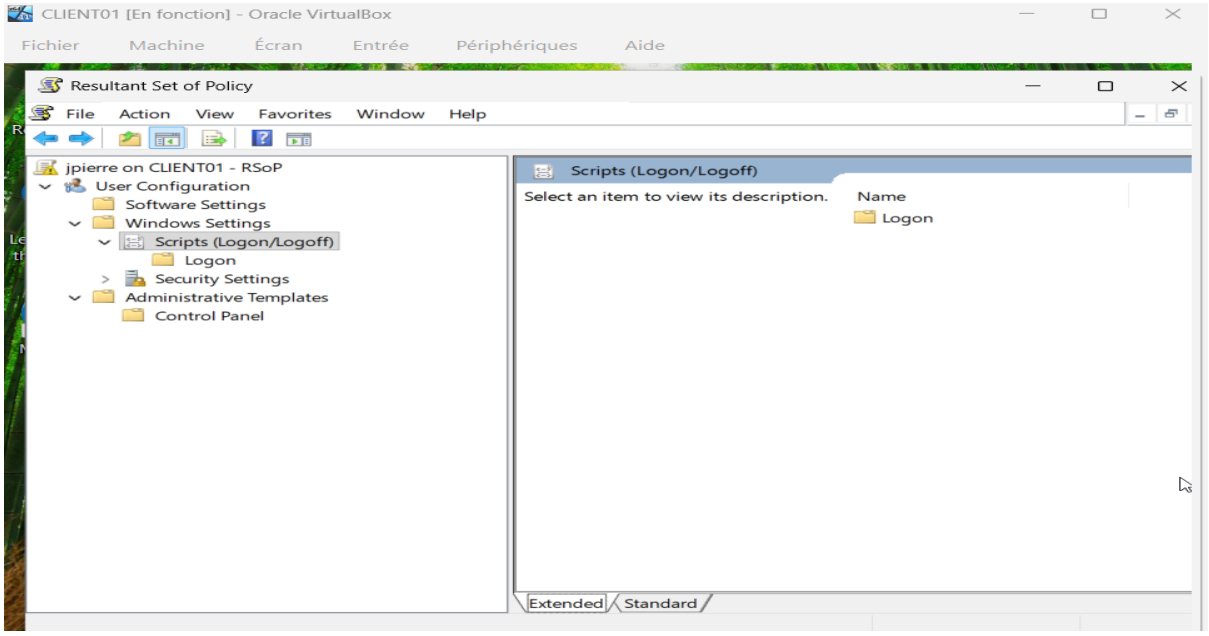
PS C:\WINDOWS\system32> _
```

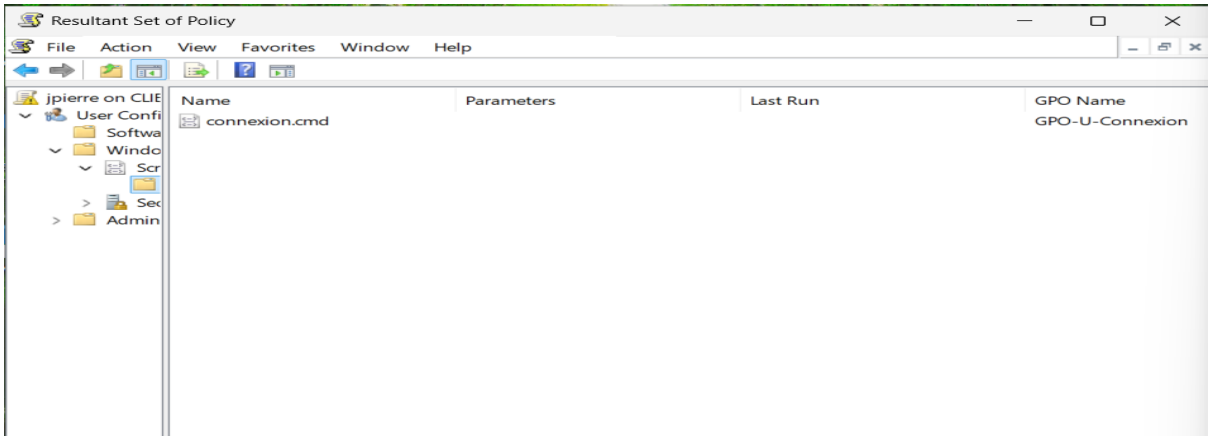

CLIENT01 [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
Welcome to Microsoft Edge, the best-performing browser on Windows
Let's start by signing you in and bringing over your passwords, history, and more.
Sign in to sync data
Start without your data
Microsoft Privacy Statement
12:12 PM
11/9/2026
Right Ctrl

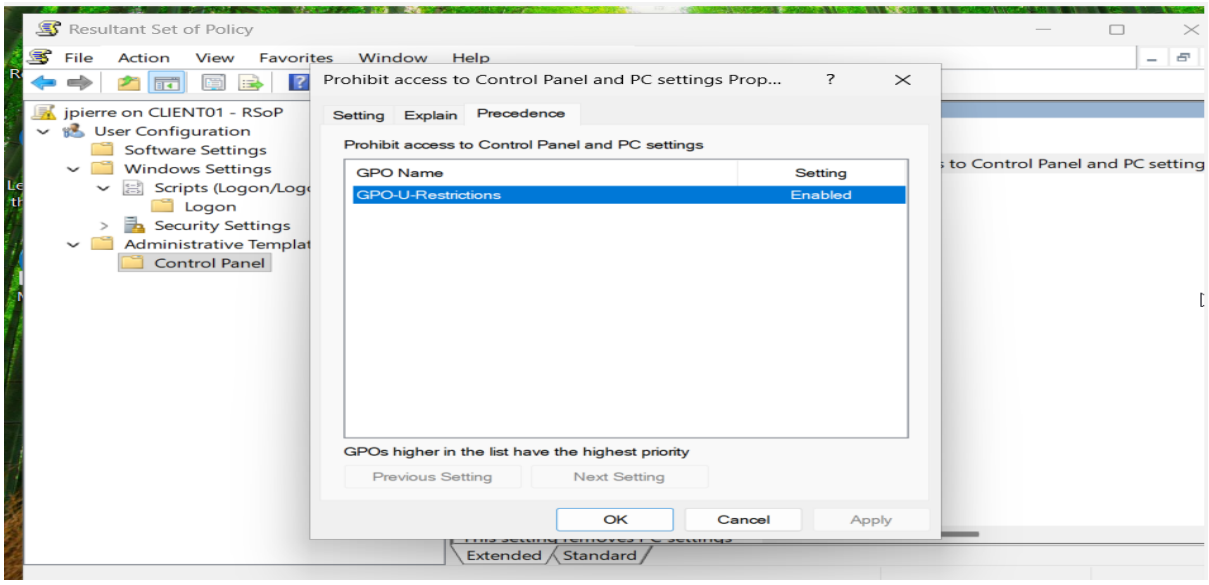

CLIENT01 [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
LAKOU\jpierre
Customize your browser
explore.microsoft.com
File C:/Users/jpierre/AppData/Local/Temp/Rapport-GPO-Lakou.html
Chat
Group Policy Results
LAKOU\jpierre
Data collected on: 11/9/2026 12:10:11 PM
show all
Summary
hide
No data available.
During last user policy refresh on 11/9/2026 10:48:44 AM
A fast link was detected More information...
Computer Details
hide
No data available.
User Details
hide
General
hide
User name LAKOU\jpierre
Domain lakou.test
Organizational Unit lakou.test/LaKouServices/Utilisateurs/Informatique
Security Group Membership show
Component Status
hide
Component Name Status Time Taken Last Process Time Event Log
Group Policy Infrastructure Success 11/9/2026 10:48:44 AM
12:13 PM
11/9/2026


CLIENT01 [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
Recycle Bin
Learn about this picture
Microsoft Edge
Resultant Set of Policy is being processed...
This Microsoft Management Console contains the RSoP snap-in defined below.
Starting with Microsoft Windows Vista Service Pack 1 (SP1), the Resultant Set of Policies (RSoP) report does not show all Microsoft Group Policy settings. To see the full set of Microsoft Group Policy settings applied for a computer or user, use the command-line tool gpresult.
Please wait while it is processed.
Selection Settings
Mode Logging
User name LAKOU\jpierre
Display user policy settings Yes
Computer name LAKOU\CLIENT01
Display computer policy settings Yes
Progress:


CLIENT01 [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
Resultant Set of Policy
File Action View Favorites Window Help
jpierre on CLIENT01 - RSoP
User Configuration
Software Settings
Windows Settings
Administrative Templates
jpierre on CLIENT01 - RSoP
Select an item to view its description.
Name
User Configuration
Extended Standard


CLIENT01 [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
Resultant Set of Policy
File Action View Favorites Window Help
jpierre on CLIENT01 - RSoP
User Configuration
Software Settings
Windows Settings
Scripts (Logon/Logoff)
Logon
Security Settings
Administrative Templates
Control Panel
Scripts (Logon/Logoff)
Select an item to view its description.
Name
Logon
Extended Standard


Resultant Set of Policy
File Action View Favorites Window Help
Name
Parameters
Last Run
GPO Name
connexion.cmd
GPO-U-Connexion



Question

1. **Qu'est-ce qu'une GPO ?**
   Une GPO est un ensemble de paramètres de configuration (sécurité, restrictions, scripts) stocké dans Active Directory. Elle s'applique automatiquement aux utilisateurs et aux ordinateurs du domaine.
2. **Pourquoi faut-il lier une GPO ?**
   Une GPO n'agit que si elle est liée à un site, un domaine ou une OU. Le lien définit sa portée : sans lien, elle existe mais ne s'applique à personne.
3. **Quelle différence entre configuration utilisateur et ordinateur ?**
   La configuration utilisateur suit le compte et s'applique à l'ouverture de session (Panneau de configuration, script de connexion). La configuration ordinateur s'applique au démarrage de la machine, quel que soit l'utilisateur (pare-feu, audit, verrouillage).
4. **Cette restriction constitue-t-elle à elle seule une protection complète du poste ?**
   Non. Elle masque seulement certaines interfaces. Un utilisateur peut encore agir autrement (ligne de commande, droits élevés, accès physique). Il faut la combiner avec le pare-feu, des comptes à privilèges limités, des mises à jour et l'audit.
5. **Pourquoi configurer les permissions de partage et NTFS ?**
   Les deux s'appliquent ensemble et la plus restrictive l'emporte. Le partage contrôle l'accès par le réseau, tandis que NTFS protège aussi l'accès local et permet un contrôle plus fin. C'est une défense en profondeur.

6. **Pourquoi utiliser un chemin UNC ?**
   Le chemin UNC (\\SRV-AD01.lakou.test\Commun) désigne la ressource directement, quel que soit le poste ou la lettre de lecteur. Le nom DNS reste valide même si l'adresse IP change.
7. **Pourquoi ne pas enregistrer un mot de passe dans le script ?**
   Le script est stocké dans le SYSVOL, lisible par les utilisateurs authentifiés : le mot de passe serait exposé. Le script utilise plutôt l'identité de la session ouverte, ce qui est plus sûr.
8. **Pourquoi ces paramètres sont-ils liés à l'OU Ordinateurs ?**
   Ce sont des paramètres de configuration ordinateur : ils ne s'appliquent qu'aux comptes machines de l'OU liée. Comme cette OU ne contient pas le contrôleur de domaine, le serveur n'est pas modifié par erreur.
9. **L'audit empêche-t-il une intrusion ?**
   Non. L'audit détecte et garde une trace (événements 4624, 4625, 4634), mais il ne bloque rien. Il sert à analyser après coup et doit être complété par des mesures préventives.
10. **Pourquoi l'heure des machines est-elle importante ?**
    Kerberos refuse l'authentification si l'écart dépasse 5 minutes. Des heures cohérentes permettent aussi de comparer les journaux de plusieurs machines. C'est ce que j'ai constaté avec CLIENT01, dont l'horloge était en retard de deux jours.